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Impaired excitability of fast-spiking neurons in a novel mouse model of *KCNC1* epileptic encephalopathy

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eLife Assessment

This study provides **important** evidence for the mechanism underlying *KCNC1*-related developmental and epileptic encephalopathy. The authors have generated and characterized a new knock-in mouse with a pathogenic mutation found in patients to determine the synaptic and circuit mechanisms contributing to *KCNC1*-associated epilepsy. They provide **convincing** evidence for reduced excitability of parvalbumin-positive fast-spiking interneurons, but not in neighboring excitatory neurons, and suggest that this may contribute to seizures and premature death in the mice.

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Abstract

The recurrent pathogenic variant *KCNC1*-p.Ala421Val (A421V) is a cause of developmental and epileptic encephalopathy characterized by moderate-to-severe developmental delay/intellectual disability, and infantile-onset treatment-resistant epilepsy with multiple seizure types including myoclonic seizures. Yet, the mechanistic basis of this disease, and of the *KCNC1* disease spectrum, is unclear. *KCNC1* encodes Kv3.1, a voltage-gated potassium channel subunit that is highly and selectively expressed in neurons capable of generating action potentials at high frequency, including parvalbumin-positive fast-spiking GABAergic inhibitory interneurons in cerebral cortex (PV-INs) known to be important for cognitive function and plasticity as well as control of network excitation to prevent seizures. In this study, we generate a novel transgenic mouse model with conditional expression of the Ala421Val pathogenic missense variant (*Kcnc1*-A421V/+ mice) to explore the specific physiological mechanisms of *KCNC1* developmental and epileptic encephalopathy. Our results indicate that global heterozygous expression of the A421V variant leads to cognitive impairment, epilepsy, and premature lethality. We observe decreased PV-IN cell surface expression of Kv3.1 via immunohistochemistry, decreased voltage-gated potassium current density in PV-INs using outside-out nucleated macropatch recordings in brain slice, and

profound impairments in the intrinsic excitability of cerebral cortex PV-INs (but not excitatory neurons) in current-clamp electrophysiology. *In vivo* two-photon calcium imaging revealed altered basal activity in *Kcnc1*-A421V/+ PV-INs and excitatory cells, as well as hypersynchronous discharges correlated with brief paroxysmal movements that were subsequently shown to be myoclonic seizures on electroencephalography. We found alterations in PV-IN-mediated inhibitory neurotransmission in young adult but not juvenile *Kcnc1*-A421V/+ mice relative to wild-type controls. Together, these results establish the specific impact of the recurrent Kv3.1-A421V variant on neuronal excitability and synaptic physiology across development to drive network dysfunction underlying *KCNC1* epileptic encephalopathy.

Introduction

Variants in *KCNC1*, which encodes the voltage-gated potassium (K⁺) channel subunit Kv3.1, cause *KCNC1*-related neurological disorders, a spectrum of clinical phenotypes ranging from nonspecific intellectual disability to progressive myoclonus epilepsy and developmental and epileptic encephalopathy (Oliver et al., 2017 [↗](#); Cameron et al., 2019 [↗](#); Park et al., 2019 [↗](#); Li et al., 2021 [↗](#); Clatot et al., 2023 [↗](#); Feng et al., 2024 [↗](#)). Kv3.1 is one of four members (Kv3.1-Kv3.4) of the Kv3 subfamily of voltage-gated K⁺ channels. Kv3 channels show unique biophysical properties relative to other voltage-gated K⁺ channels including a depolarized voltage-dependence of activation, fast rates of activation and deactivation, and little/no inactivation, properties that are exquisitely tuned to generate brief spikes and limit inter-spike interval, and thereby support rapid cycling required for reliable fast-spiking in Kv3-expressing neurons (Weiser et al., 1995 [↗](#); Massengill et al., 1997 [↗](#); Sekirnjak et al., 1997 [↗](#); Gan and Kaczmarek, 1998 [↗](#); Martina et al., 1998 [↗](#); Wang et al., 1998 [↗](#); Erisir et al., 1999 [↗](#); Rudy and McBain, 2001 [↗](#); Akemann and Knöpfel, 2006 [↗](#); Sacco et al., 2006 [↗](#); Martina et al., 2007 [↗](#)). Thus, Kv3 channels are highly and specifically expressed in cellular populations throughout the brain known to generate APs at high-frequency, including cerebellar granule and Purkinje cells, neurons of the reticular thalamus, as well as parvalbumin-positive fast-spiking GABAergic inhibitory interneurons (PV-INs) in the neocortex, hippocampus and basal ganglia (Rudy et al., 1999 [↗](#); Kaczmarek and Zhang, 2017 [↗](#)). Alterations in Kv3.1 function would be expected to have a profound impact on neuronal excitability of fast-spiking neurons with downstream effects on circuits containing Kv3-expressing cells.

Our previous study using a novel mouse model of Progressive Myoclonus Epilepsy Type 7 (PME7 or EPM7) harboring the recurrent missense variant *KCNC1*-p.Arg320His indicated that loss of Kv3.1 function alters excitability and synaptic neurotransmission in cerebral cortex PV-INs and cerebellar granule cells in adult heterozygous *Kcnc1*-p.R320H/+ mice (Feng et al., 2024 [↗](#)). In contrast to EPM7, patients harboring de novo heterozygous *KCNC1*-p.Ala421Val variants exhibit developmental and epileptic encephalopathy (DEE), with moderate to severe developmental delay/intellectual disability without regression, variable but mild nonprogressive ataxia, and treatment-resistant epilepsy onset in infancy with multiple seizure types including myoclonic seizures (Oliver et al., 2017 [↗](#); Cameron et al., 2019 [↗](#); Park et al., 2019 [↗](#); Li et al., 2021 [↗](#)). Examination of the function of voltage-gated K⁺ channels containing variant versus wild-type (WT) Kv3.1 in heterologous systems has indicated that the A421V variant is a near-complete loss-of-function at the level of the channel, generating K⁺ currents that are significantly reduced in magnitude relative to WT (Cameron et al., 2019 [↗](#); Park et al., 2019 [↗](#)). Hence, while both the R320H and A421V variants are loss of function with a proposed dominant negative action on tetrameric Kv3 channels composed of WT and variant subunits in heterologous systems, the A421V variant is a markedly more severe loss of function, consistent with the associated clinical phenotype with earlier age of onset and treatment-resistant epilepsy. The A421 residue is localized between the selectivity filter and the PVP motif of Kv3.1. Molecular modeling shows that the A421V variant does not lead to obvious steric hindrance in the channel yet could possibly influence gating and selectivity through the addition of hydrophobic carbon atoms in the Kv3.1 pore (Li et al., 2021 [↗](#)). Yet, the precise mechanisms underlying how the A421V variant impacts

native neuronal Kv3 currents, neuronal physiology, and ultimately results in developmental and epileptic encephalopathy, and how this differs from other disease-associated variants in *KCNC1*, is unclear.

In this study, we generated a novel mouse model of *KCNC1* DEE – *Kcnc1*-Flox(p.Ala421Val)/+ (i.e., *Kcnc1*-A421V/+ mice – to determine the impact of heterozygous expression of the *Kcnc1*-p.A421V variant as seen in patients on native voltage-gated K⁺ channel currents, intrinsic excitability of Kv3.1-expressing neurons, inhibitory synaptic neurotransmission and function in cortical microcircuits, and epilepsy phenotype. Our results indicate that global expression of the *Kcnc1*-p.A421V allele results in developmental impairment, cognitive dysfunction, epilepsy including prominent myoclonic seizures, and premature lethality due to seizure-induced sudden death. Patch-clamp electrophysiological recordings demonstrate that Kv3-like voltage-gated K⁺ current density is significantly reduced in PV-INs driven at least in part by impaired trafficking and cell surface expression of Kv3.1, with resulting alterations in AP waveform and impaired intrinsic excitability. Excitatory cell physiology was unchanged in the *Kcnc1*-A421V/+ mice, suggesting that the phenotype is related to inhibitory neuron dysfunction. Investigation of synaptic neurotransmission revealed no significant differences between WT and *Kcnc1*-A421V/+ PV-IN-mediated inhibitory neurotransmission at the early juvenile time window (P16–21), but significantly altered properties at the young adult time point (P32–42), consistent with the observed progressive worsening of epilepsy in the mouse model and suggesting that altered Kv3.1 function leads to impairments in PV-IN synaptic function in a developmentally-regulated manner. Overall, these results indicate that the *Kcnc1*-A421V variant is physiologically loss-of-function in native neurons with resulting impairment of intrinsic excitability and synaptic transmission of Kv3.1-expressing parvalbumin-positive fast-spiking cells yielding epilepsy and cognitive impairment.

Results

Generation of the *Kcnc1*-A421V/+ mouse model of *KCNC1* Epilepsy

We generated a novel transgenic mouse (see Materials and Methods) that conditionally expresses *Kcnc1*-p.A421V/+ (*Kcnc1*-p.A421V/+ mice) homologous to a recurrent *KCNC1* variant previously identified in human patients with developmental and epileptic encephalopathy (Oliver et al., 2017 [↗](#); Cameron et al., 2019 [↗](#); Park et al., 2019 [↗](#)). Briefly, the *Kcnc1* c.1262C>T missense was introduced into an ES cell line via gene targeting, converting a GCT to GTT and leading to the Ala421Val amino acid change. A targeting vector containing part of intron 1 followed by the coding sequence of exons 2–4 flanked by loxP sites was then introduced upstream of the modified endogenous sequence (fig. 1A [↗](#)). Thus, in the absence of Cre recombinase, there is expression of the introduced 5' wild-type exons 2–4; in the presence of Cre recombinase, there is Cre-mediated excision of the floxed wild-type exons 2–4 coding sequence and expression of *Kcnc1* harboring the c.1262C>T substitution resulting in the single amino acid change p.Ala421Val (fig. 1A [↗](#)). Sanger sequencing confirmed the knock-in point mutation in exon 2 and subsequent PCR showed Cre-dependent genome recombination of the variant (fig. 1B–C [↗](#)). We utilized a breeding strategy which allowed us to examine the behavior and physiology of mice expressing the *Kcnc1* variant globally (via cross to ActB-Cre mice; JAX#: 003376), as is presumed the case with human patients harboring *KCNC1*.A421V as a de novo pathogenic variant (fig. 1D [↗](#)). We also used a transgenic mouse line (C57BL/6-Tg(Pvalb-tdTomato)15Gfng/J; JAX#: 027395) which fluorescently labels parvalbumin-positive inhibitory GABAergic interneurons (PVINs) with the red fluorescent protein tdTomato driven by the endogenous parvalbumin promoter (fig. 1D [↗](#)). Our triple transgenic breeding strategy therefore produced experimental *Kcnc1*-A421V/+ mice and WT littermates of both sexes containing Cre and with 50% harboring the tdTomato allele to guide physiological experiments targeting PV-INs. The overall survival curve demonstrated that *Kcnc1*-A421V/+ mice (*N*=33) exhibited premature death relative to their WT littermates (*N*=46) with no *Kcnc1*-A421V/+ mice surviving beyond 122 days (***P*<0.001 by Mantel-Cox Test; fig. 1E [↗](#)).

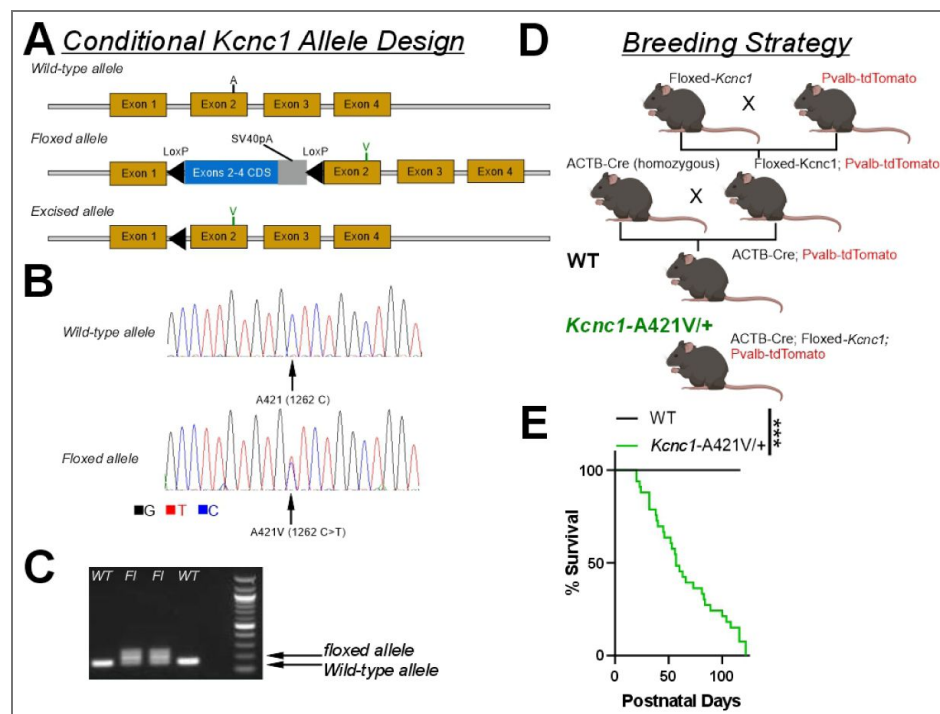


Figure 1. Design of a novel mouse model of *Kcnc1* developmental and epileptic encephalopathy.

A. Design and structure of the conditional *Kcnc1*-A421V allele. Upon Cre-mediated recombination, the inserted WT coding sequence (CDS) flanked by LoxP sites is removed and the A421V variant inserted into exon 2 is expressed. **B.** Sequencing results indicate successful targeting of c.1262C>T to introduce the heterozygous A421V variant. **C.** PCR confirmation of two HET founders (*Kcnc1*-A421V/+) and two WT littermates. 167 bp, WT allele fragment; 207 bp, floxed allele fragment. **D.** Breeding strategy to generate control and experimental mice in which the *Kcnc1*-A421V variant is expressed globally and PV cells are fluorescently-labeled for targeted recording. **E.** Survival plot of WT ($N=46$; black) and *Kcnc1*-A421V/+ ($N=33$; green) mice. *** $P<0.001$ by log-rank Mantel-Cox curve comparison.

Counts of PV cells per unit of cortical area were not different between WT and *Kcnc1*-A421V/+ mice (P24-33) indicating that expression of A421V did not alter the density of cortical PV-INs (Figure 1—figure supplement 1A-B), consistent with the fact that interneuron migration is complete prior to expression of *Kcnc1* in mouse. We separately used immunohistochemistry to validate our genetic strategy for labeling PV-INs (Figure 1—figure supplement 1C-D): The average sensitivity rates were greater than 0.8 in both groups as previously reported (Kaiser et al., 2016) and were not different by genotype (Figure 1—figure supplement 1C), with the average false positive identification rate less than 0.1 for each group (Figure 1—figure supplement 1D).

Behavioral Testing of *Kcnc1*-A421V/+ Mice

Kcnc1-A421V/+ mice underwent an assessment of developmental milestones at postnatal day five through 15 (P5-15), as done previously (Feng et al., 2024). Although *Kcnc1*-A421V/+ mice exhibited reduced body (Figure 1—figure supplement 2A-B) and brain (Figure 1—figure supplement 2C) weights relative to their WT littermates – as seen previously with *Kcnc1* knockout mice (Ho et al., 1997) – we did not detect other developmental abnormalities in the onset of fur appearance, eye opening, ear canal opening, incisor eruption, head elevation, shoulder elevation, auditory startle, horizontal screen test, vertical screen test, cliff avoidance, quadruple walking, and negative geotaxis (Figure 1—figure supplement 2D-E). These results suggest that while body and brain weights are reduced, the *Kcnc1*-A421V/+ mice show otherwise typical gross anatomical and functional development within the early developmental timepoint examined (P5-15), as determined by the tests readily available for such evaluation.

We next tested whether cognitive function was altered in adult *Kcnc1*-A421V/+ mice (P35-65). To this end, we assessed spatial memory in juvenile *Kcnc1*-A421V/+ mice in the Barnes Maze task (Figure 2A). *Kcnc1*-A421V/+ mice showed a significant delay compared to controls in the acquisition of the escape hole position (Figure 2A-B). This defect was most pronounced by significantly longer escape latencies during the second day of acquisition trials, suggesting an impairment in spatial learning. During the probe trial, conducted in the absence of the escape box (to assess memory retention), *Kcnc1*-A421V/+ mice spent the same time as wild type littermates in the target quadrant (fig. 2C), indicating intact long-term memory of the escape hole position.

We then assessed spatial working memory using the Y maze spontaneous alternation test (fig. 2D-F). Spontaneous alternation is a behavior driven by the innate tendency of rodents to alternate between recently visited arms to explore previously unvisited areas of the maze. Relative to WT littermates, *Kcnc1*-A421V/+ displayed a statistically significant decrease in the percentage of spontaneous alternations (fig. 2D). Importantly, the total number of arm entries (fig. 2E) and the distance travelled (fig. 2F) did not differ between genotypes, suggesting that the observed deficit was not due to differences in general activity, motor function, or exploratory drive. Taken together, these behavioral observations indicate deficits in both spatial learning and working memory systems in *Kcnc1*-A421V/+ mice.

Kcnc1-A421V/+ mice exhibit reduced voltage-gated potassium channel currents and altered Kv3.1 expression

Previous studies in heterologous expression systems have reported that the A421V variant is physiologically loss-of-function and generates strongly attenuated voltage-gated K⁺ channel currents in *Xenopus laevis* oocytes (Cameron et al., 2019; Park et al., 2019), albeit with conflicting conclusions related to the presence of a dominant-negative action. In our recordings of HEK cells expressing WT and A421V Kv3.1 subunits, A421V was profoundly loss of function although a small magnitude K⁺ current was detectable (which cannot be easily distinguished from the small endogenous delayed rectified potassium current present in HEK cells; Figure 3—figure supplement 1). A 50:50 mixture of WT and A421V subunits, approximating the clinical condition, produced K⁺ currents that were 42% of the magnitude and slightly shifted in the hyperpolarized direction relative to the WT condition (Figure 3—figure supplement 1D-E). Considering that Kv3.1 channels form heterotetramers with other Kv3 channel isoforms (likely Kv3.2) in cortical PV-INs, we also sought to clarify the impact of the A421V *Kcnc1* variant on native neuronal

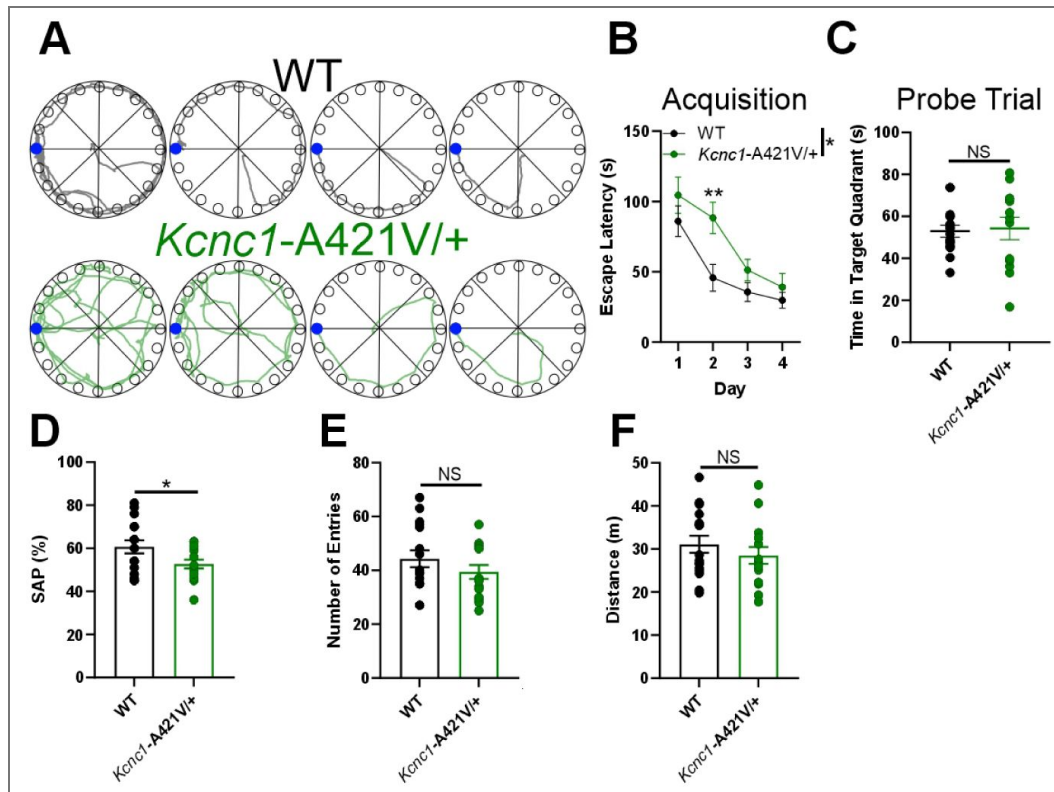


Figure 2. Impaired Cognitive Function in *Kcnc1-A421V/+* Mice.

A. Example pathways for WT (black) and *Kcnc1-A421V/+* (green) during the four-day acquisition phase of the Barnes maze test. Blue dot indicates the escape hole. **B.** Average escape latency during the acquisition phase for WT (black; N=15) and *Kcnc1-A421V/+* (green; N=13) mice. **C.** Group data for time spent in target quadrant during probe trial of Barnes maze. **D-F.** Group data for WT (black; N=16) and *Kcnc1-A421V/+* mice (green; N=15) during the Y-maze test for spatial working memory. **D.** Spontaneous alternation percentage. **E.** Total number of arm entries. **F.** Total distance traveled. Data are shown as mean $\pm$ SEM and were analyzed by Two-way repeated measures ANOVA with Tukey's post-hoc test (B) and unpaired t-test (D-F). Significance is denoted as * $P < 0.05$ or ** $P < 0.01$.

voltage-gated K^+ channel function in neocortical layer II-IV PV-INs, known to express high levels of Kv3.1 (Chow et al., 1999 [\[1\]](#)), from *Kcnc1*-A421V/+ mice at the juvenile timepoint of P16-21. Outside-out nucleated macropatch recordings allowed for high-quality electrophysiological recordings of somatic voltage-gated K^+ currents (fig. 3A-C [\[2\]](#)). Relative to WT controls ($n=13$ cells, $N=3$ mice), the current density of voltage-gated K^+ currents in the *Kcnc1*-A421V/+ mice ($n=17$ cells, $N=3$ mice) was markedly reduced across all voltages examined ($***P<0.001$; Repeated-measures Two-way ANOVA; fig. 3B-D [\[2\]](#)). Peak current densities were significantly lower in PV-INs from *Kcnc1*-A421V/+ mice compared to WT controls ($***P<0.001$; t-test; fig. 3E [\[2\]](#)). We did not observe differences in the voltage-dependence (fig. 2F [\[2\]](#)) or kinetics of activation (fig. 3G [\[2\]](#)) when comparing voltage-gated K^+ channel currents from WT and *Kcnc1*-A421V mice. Together, these results demonstrate that PV-IN voltage-gated K^+ channel function is strongly impaired in the context of the *Kcnc1*-p.A421V variant.

The markedly decreased K^+ current magnitude observed in PV-INs without apparent alterations in gating properties is consistent with impaired conductance of the population of Kv3 channels, but could also be explained by impaired trafficking to the cell membrane. To investigate this possibility, we examined Kv3.1 expression in PV-INs from young adult (P24-33) WT and *Kcnc1*-A421V/+ mice via immunohistochemistry. Our results suggested that the amount of Kv3.1 at the membrane relative to cytosol was significantly altered in the *Kcnc1*-A421V/+ mice ($n=48$ cells, $N=5$ mice) compared to WT controls ($n=49$ cells, $N=6$ mice; fig. 3H-I [\[2\]](#)). These findings support the conclusion that impaired trafficking to the cell surface at least contributes to the observed decrease in K^+ current in PV-INs in *Kcnc1*-A421V/+ mice.

Intrinsic excitability of PV-INs is altered in *Kcnc1*-A421V/+ mice

We next examined intrinsic neuronal excitability in PV-INs in somatosensory neocortex layers II-IV to determine the impact of the voltage-gated K^+ channel dysfunction across two age ranges, juvenile (P16-21; fig. 4A-E [\[2\]](#)) and young adult mice (P32-42; fig. 4F-J [\[2\]](#)). We performed whole-cell current-clamp recordings to generate a detailed comparison of passive membrane properties, properties of individual APs, and of repetitive firing, for PV-INs from primary somatosensory neocortex in WT vs. *Kcnc1*-A421V/+ mice at both time points (fig. 4 [\[2\]](#) and tables 1 [\[2\]](#) and 2 [\[2\]](#)). PV-INs from both genotypes generated trains of repetitive APs in response to depolarizing current injection; however, frequency was profoundly reduced in *Kcnc1*-A421V/+ mice relative to the WT control PV-INs at all current magnitudes examined ($***P<0.001$, Repeated-measures Two-way ANOVA; fig. 4B-D [\[2\]](#)) regardless of whether resting membrane potential was normalized across cells with DC bias current (Figure 4—figure supplement 1 [\[2\]](#)). When examining the properties of individual APs, *Kcnc1*-A421V/+ and WT PV-INs showed significant differences in downstroke velocity and half-maximal AP duration (APD50), properties that are determined by Kv3 channel function. AP amplitude was elevated in the *Kcnc1*-A421V/+ mice at juvenile (P16-21; fig. 4E [\[2\]](#); table 1 [\[2\]](#)) and adult (P32-42; fig. 4J [\[2\]](#); table 2 [\[2\]](#)) timepoints, but only reached significance at P32-42 ($***P<0.001$, unpaired t-test). Overall, PV-INs from *Kcnc1*-A421V/+ mice exhibited specific impairments consistent with reduction in Kv3 current including altered properties of individual APs leading to a reduction in firing frequency, with a relative preservation of passive membrane properties not thought to be directly regulated by Kv3 channels.

We sought to further extend these results by examining other cell population linked to epilepsy pathogenesis. Recordings of neocortical layer V PV-INs from juvenile (P16-21) *Kcnc1*-A421V/+ mice exhibited more subtle abnormalities compared to littermate WT control PV-INs, showing reduction in AP frequency only at the largest current injection magnitudes ($***P<0.001$ for interaction between genotype and current injection, Figure 4—figure supplement 2 [\[2\]](#)). These results are consistent with previous reports that Kv3.1 comprises a relatively lower proportion of the overall Kv3 expression in deeper layer cortical PV-INs due to higher relative levels of Kv3.2 expression (Chow et al., 1999 [\[1\]](#)).

We also examined PV-positive neurons in the nucleus of the reticular thalamus (RTN), which predominantly express Kv3.1 and Kv3.3 (Porcello et al., 2002 [\[3\]](#); Espinosa et al., 2008 [\[4\]](#)). In response to hyperpolarizing current injections of various magnitudes, RTN neurons from *Kcnc1*-

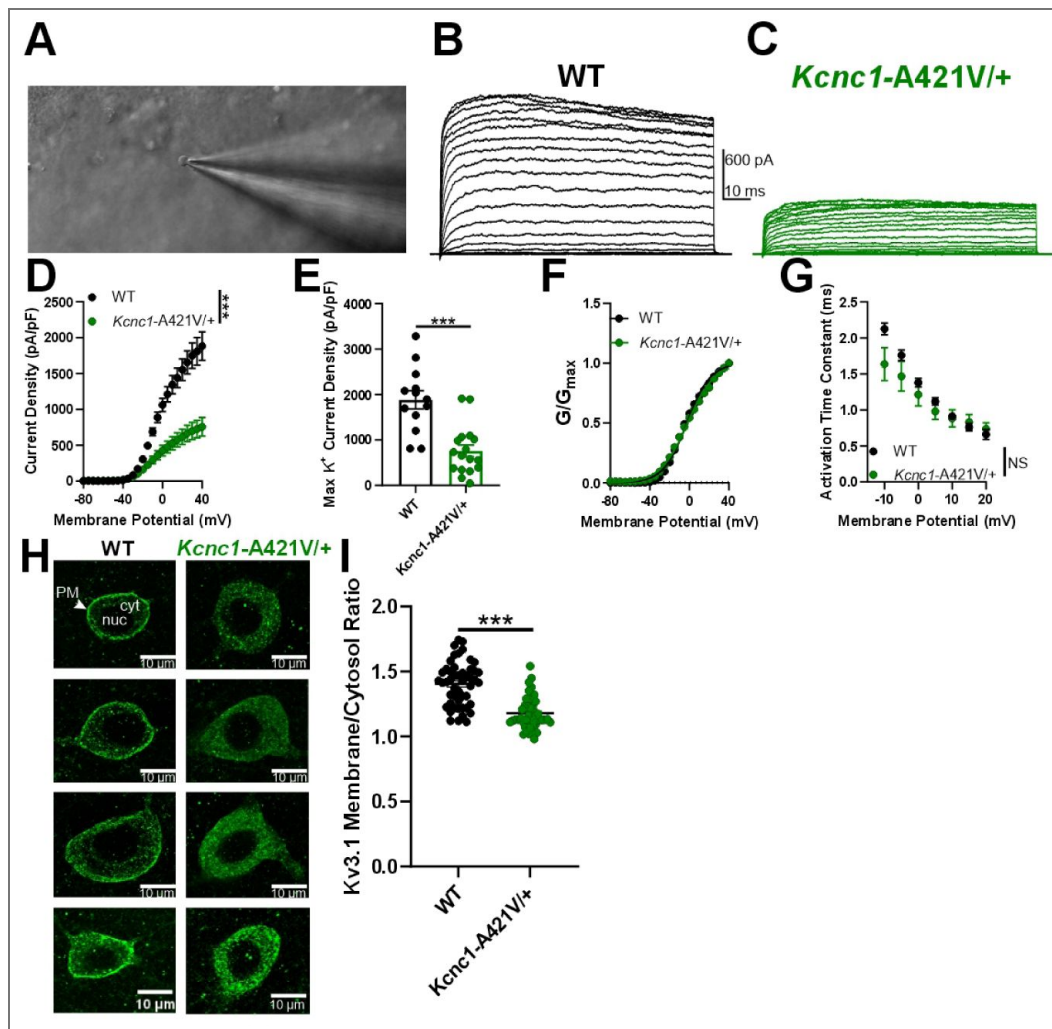


Figure 3. PV-INs from *Kcnc1-A421V/+* mice exhibit attenuated voltage-gated potassium channel currents and impaired membrane Kv3.1 expression.

A. Representative image of a cell being recorded in the outside-out nucleated macropatch configuration. **B-C.** Example family of traces of voltage-gated K^+ channel currents from a PV-IN from WT (**B**, black) and *Kcnc1-A421V/+* (**C**, green) mice (P16-21). **D.** Average voltage-gated K^+ channel current density for WT ($n=13$ macropatches, $N=3$ mice) and *Kcnc1-A421V/+* ($n=17$, $N=3$ mice). **E.** Maximum K^+ channel current density per PV-IN macropatch in WT and *Kcnc1-A421V/+* mice. **F.** Averaged normalized plots of K^+ conductance relative to voltage command indicating voltage-dependence of activation curves for WT and *Kcnc1-A421V/+* mice. **G.** Average activation time constant for the voltage-gated K^+ channel currents relative to voltage command potential in both WT and *Kcnc1-A421V/+* mice. **H.** Representative images of individual cortical PV-INs from WT and *Kcnc1-A421V/+* mice stained for Kv3.1 (green). Plasma membrane (PM), nucleus (nuc), and cytosol (cyt) are indicated in the top left panel. Note the markedly less pronounced Kv3.1 intensity in the plasma membrane in each of the *Kcnc1-A421V/+* examples and more prominent cytosolic labeling (presumably corresponding to endoplasmic reticulum). **I.** Group quantification of ratio of membrane to cytosolic Kv3.1 for WT ($n=49$ cells, $N=6$ mice) and *Kcnc1-A421V/+* ($n=48$ cells, $N=5$ mice). Data are shown as mean $\pm$ SEM or individual data points and significance was determined using either Repeated-measures Two-way ANOVA or unpaired t-test where *** $P<0.001$.

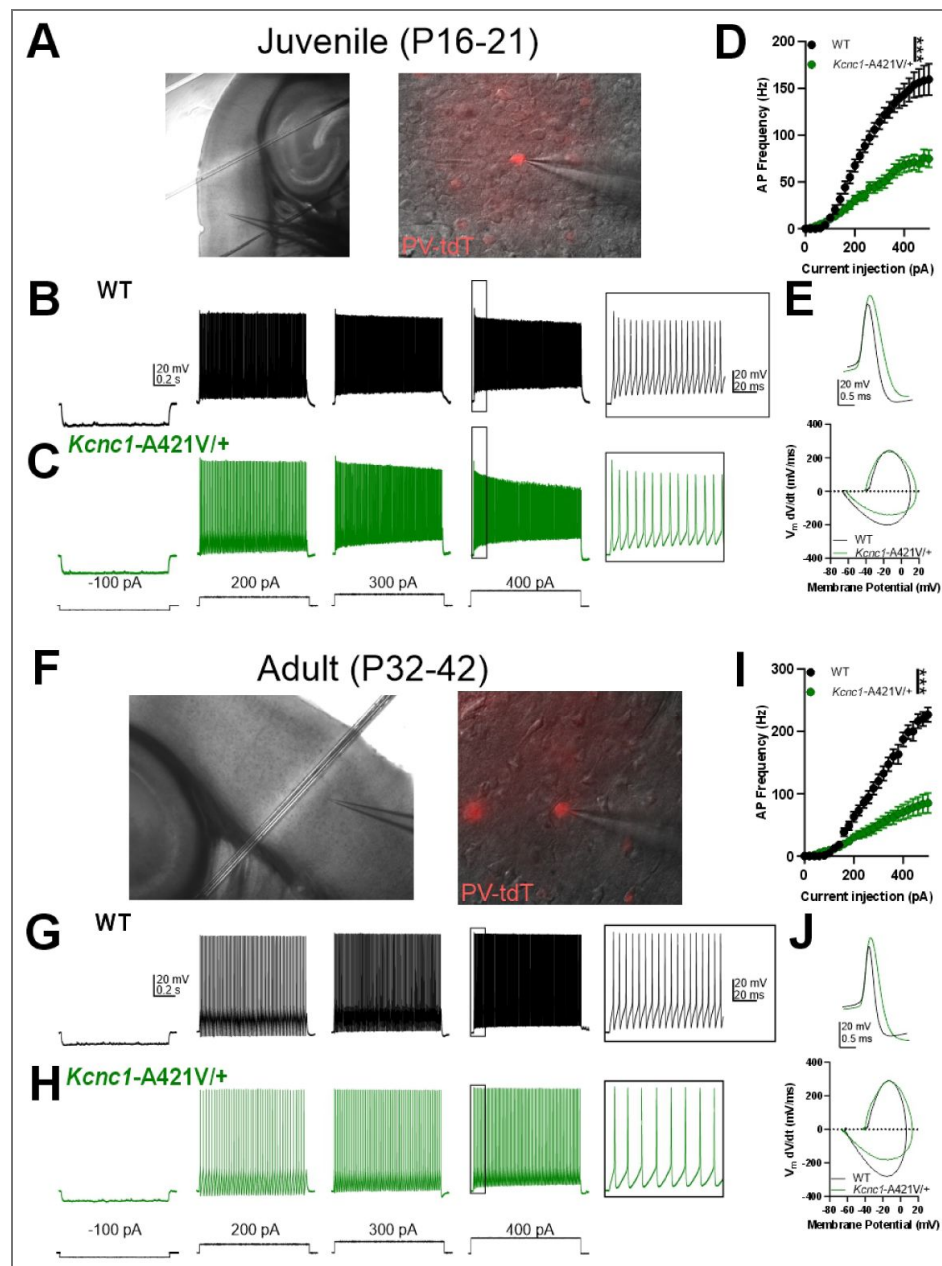


Figure 4. Impaired PV-IN intrinsic Excitability in Juvenile and Adult *Kcnc1-A421V/+* mice.

A. Representative images taken at 10X (left) and 40X (right) magnification of a layer IV neocortical PV-IN recorded in the whole-cell configuration to characterize intrinsic excitability. **B-C.** Representative example traces for juvenile (P16-21) WT (B, black) and *Kcnc1-A421V/+* (C, green) PV-INs generating APs at current injections of -100, 200, 300, and 400 pA. The inset shows an expanded view of APs generated in response to the 400-pA current step in both genotypes. **D.** Average relationship between PV-IN AP frequency in response to a range of current injections for juvenile WT ($n=20$ cells, $N=9$ mice) and *Kcnc1-A421V/+* ($n=36$ cells, $N=12$ mice). **E.** Representative overlaid examples of single APs and the corresponding phase plots for juvenile WT (black) and *Kcnc1-A421V/+* (green) PV-INs. **F.** Representative images for a layer IV neocortical PV-IN from an adult mouse. **G-H.** Representative example traces displaying intrinsic excitability in adult (P32-42) WT and *Kcnc1-A421V/+* PV-INs. Inset show an expanded view of APs induced by the 400-pA current step. **I.** Average relationship between PV-IN AP frequency and current injection for adult WT ($n=14$ cells, $N=3$ mice) and *Kcnc1-A421V/+* mice ($n=17$ cells, $N=5$ mice). **J.** Representative overlaid examples of single APs and the corresponding phase plots for adult WT (black) and *Kcnc1-A421V/+* (green) PV-INs. Data are shown as mean $\pm$ SEM and significance (***) $P < 0.001$ was determined using repeated-measures Two-way ANOVA.

Table 1. Membrane and Action Potential Properties of WT and *Kcnc1*-A421V/+ Neurons at P16-21.

Cell Type	Group	V _m (mV)	AP Threshold (mV)	Upstroke Velocity (mV/ms)	Downstroke Velocity (mV/ms)	AP Amplitude (mV)	APD50 (ms)	Input Resistance (MΩ)	Rheobase (pA)	AHP (mV)
Layer II-IV PV-INs	WT (N=20,9)	-72.3 ± 1.3	-39.8 ± 0.8	253 ± 8	-183 ± 9	52.1 ± 1.8	0.40 ± 0.02	142 ± 13	93 ± 9	-63.7 ± 1.0
	<i>Kcnc1</i> -A421V/+ (N=36, 12)	-67.4 ± 1.3	-41.6 ± 0.5	252 ± 9	-138 ± 8	57.1 ± 1.6	0.54 ± 0.03	170 ± 21	145 ± 19	-63.0 ± 0.8
	Statistical Comparison	*P=0.017	P=0.065	P=0.96	**P=0.0012	P=0.055	**P=0.0053	P=0.29	P=0.19	P=0.62
Layer IV Exc. Cells	WT (N=23, 3)	-66.7 ± 0.7	-43.4 ± 0.7	303 ± 13	-68.6 ± 3.9	79.8 ± 1.3	1.06 ± 0.05	155 ± 15	34 ± 6	-60.2 ± 0.5
	<i>Kcnc1</i> -A421V/+ (N=22,3)	-67.6 ± 1.0	-41.8 ± 0.7	286 ± 15	-64.5 ± 3.7	77.8 ± 1.6	1.10 ± 0.04	159 ± 14	37 ± 6	-60.8 ± 0.6
	Statistical Comparison	P=0.49	P=0.11	P=0.38	P=0.45	P=0.35	P=0.46	P=0.86	P=0.69	P=0.47
Layer V PV-INs	WT (N=15, 3)	-65.9 ± 1.2	-38.6 ± 1.0	309 ± 22	-223 ± 17	57.5 ± 1.9	0.37 ± 0.02	147 ± 13	85 ± 11	-65.8 ± 1.0
	<i>Kcnc1</i> -A421V/+ (N=12, 3)	-66.3 ± 1.7	-40.4 ± 1.0	278 ± 26	-164 ± 15	59.0 ± 2.1	0.47 ± 0.03	141 ± 17	97 ± 13	-65.7 ± 1.2
	Statistical Comparison	P=0.82	P=0.21	P=0.38	*P=0.016	P=0.60	*P=0.014	P=0.76	P=0.49	P=0.92
RTN	WT (N=18, 4)	-55.1 ± 1.4	-39.2 ± 0.8	227 ± 15	-189 ± 11	49.2 ± 1.8	0.38 ± 0.02	247 ± 39	24 ± 7	-64.5 ± 0.5
	<i>Kcnc1</i> -A421V/+ (N=19, 5)	-57.5 ± 2.3	-38.1 ± 1.0	195 ± 12	-160 ± 7	45.6 ± 1.7	0.42 ± 0.02	236 ± 30	48 ± 10	-63.7 ± 0.8
	Statistical Comparison	P=0.39	P=0.41	P=0.10	*P=0.034	P=0.15	P=0.092	P=0.82	P=0.0805	P=0.40

Table 2. Membrane and AP Properties of Adult (P32-42) WT and *Kcnc1*-A421V/+ PV-INs.

Cell Type	Group	V _m (mV)	AP Threshold (mV)	Upstroke Velocity (mV/ms)	Downstroke Velocity (mV/ms)	AP Amplitude (mV)	APD50 (ms)	Input Resistance (MΩ)	Rheobase (pA)	AHP (mV)
Layer II-IV PV-INs	WT (N=14,3)	-66.1 ± 1.4	-41.5 ± 0.9	294 ± 25	-248 ± 21	51.1 ± 2.0	0.31 ± 0.02	141 ± 11	91 ± 10	-67.6 ± 1.1
	<i>Kcnc1</i> -A421V/+ (N=17, 5)	-68.3 ± 2.3	-43.1 ± 0.6	324 ± 16	-164 ± 18	67.8 ± 2.3	0.58 ± 0.07	173 ± 25	106 ± 23	-66.0 ± 1.2
	Statistical Comparison	P=0.45	P=0.16	P=0.29	**P=0.0051	***P<0.001	**P=0.0026	P=0.28	P=0.87	P=0.34
Layer IV Exc. Cells	WT (N=12,4)	-68.6 ± 0.9	-41.7 ± 0.5	363 ± 17	-83.2 ± 5.1	83.4 ± 0.9	0.89 ± 0.04	110 ± 10	88 ± 10	-55.7 ± 1.1
	<i>Kcnc1</i> -A421V/+ (N=12, 4)	-67.2 ± 0.5	-40.0 ± 0.7	338 ± 18	-78.6 ± 4.7	81.0 ± 1.7	0.92 ± 0.03	131 ± 10	58 ± 8	-56.8 ± 0.9
	Statistical Comparison	P=0.19	P=0.057	P=0.33	P=0.52	P=0.24	P=0.69	P=0.14	*P=0.023	P=0.47

A421V/+ mice (P16-21; $N=19$ cells, 5 mice) generated fewer rebound APs than their WT counterparts ($n=16$ cells, $N=4$ mice; Figure 4—figure supplement 1B-C). As in neocortical PV-INs, the relationship between AP frequency and depolarizing current injection was attenuated in reticular thalamic cells from *Kcnc1*-A421V/+ mice relative to WT counterparts ($*P=0.0109$; Figure 4—figure supplement 3B,D). And as in neocortical PV-INs, the magnitude of the downstroke velocity was significantly reduced in *Kcnc1*-A421V/+ RTN neurons, while other parameters did not reach significance at this timepoint (Table 1). Thus, impairments in intrinsic physiology extend beyond cortical PV-INs to Kv3.1-expressing cells in other brain regions, but with cell populations known to express Kv3.2 or Kv3.3 showing more subtle impairment than PV-INs in the superficial neocortical layers.

Normal physiological function in excitatory neurons from *Kcnc1*-A421V/+ mice

We next investigated the voltage-gated K^+ channel function and intrinsic excitability in layer IV excitatory cells from both WT and *Kcnc1*-A421V/+ mice at P16-21 (fig. 5C) and P32-42 (Figure 5—figure supplement 1). While voltage-gated K^+ channel currents in neocortical layer IV excitatory neurons were of significantly lower magnitude compared to that observed in PV-INs, there were no genotype differences in K^+ currents between excitatory cells from WT and *Kcnc1*-A421V/+ mice, consistent with a lack of Kv3.1 expression in excitatory cells (fig. 5B-D). Neither voltage-dependent current density (fig. 5D), peak current density (fig. 5E), voltage-dependent activation (fig. 5F), nor the voltage-dependent rate of activation (fig. 5G) were altered in excitatory neurons of *Kcnc1*-A421V/+ mice relative to WT controls. We also recorded these cells in current-clamp to characterize intrinsic excitability (fig. 5H-K). Across a range of depolarizing current injection magnitudes, we did not detect any differences in steady-state AP frequency in excitatory neurons between WT and *Kcnc1*-A421V/+ mice (fig. 5I-K). We also did not detect any statistically significant alterations in the passive membrane properties or properties of single APs between WT and *Kcnc1*-A421V/+ mice (table 1). At the adult timepoint (P32-42), we similarly did not observe spiking abnormalities (Figure 5—figure supplement 1; table 2). Overall, these data suggest that intrinsic excitability of neocortical excitatory neurons is not altered in juvenile or adult *Kcnc1*-A421V/+ mice (either directly or via a secondary network effect), while impaired intrinsic excitability of PV-positive neurons in the neocortex and reticular thalamus is most likely a direct result of cell-autonomous reduction in Kv3.1 current density.

PV-IN and excitatory cell synaptic neurotransmission is functionally intact in juvenile *Kcnc1*-A421V/+ mice

Within fast-spiking cells, Kv3 channels are expressed in specific subcellular compartments and are functionally involved not only in AP generation, but also in AP propagation along the axon and in inhibitory neurotransmission at the synaptic terminal via regulation of synaptic action potential (AP) waveform (Goldberg et al., 2005; Rowan et al., 2014, 2016; Rowan and Christie, 2017). Additionally, prior work has shown that block of presynaptic Kv3 current leads to an increase in the efficacy of synaptic transmission albeit with enhanced short-term synaptic depression (Ishikawa et al., 2003; Brooke et al., 2004; Goldberg et al., 2005). For that reason, we sought to determine the impact of the A421V variant on PV-IN-mediated inhibitory synaptic neurotransmission in juvenile (P16-21) WT vs. *Kcnc1*-A421V/+ mice (fig. 6). We collected simultaneous whole-cell patch-clamp electrophysiology recordings from one neocortical layer II-IV PV-IN and one nearby ($<100\ \mu\text{m}$ inter-soma distance) excitatory neuron of which 21 of 64 (32.8%) WT neuron pairs and 15 of 43 (31.2%) *Kcnc1*-A421V/+ neuron pairs exhibited unitary inhibitory post-synaptic currents (uIPSCs) in the excitatory cell in response to AP generation in the PV-IN at 5, 10, 20, 40, 80, and 120Hz (fig. 6A-E). Rates of failure of the first five APs (AP is successfully initiated in the PV-IN, but no uIPSC is observed in the excitatory cell) were not different in WT vs. *Kcnc1*-A421V/+ mice (fig. 6F). The magnitudes of the first five uIPSCs at various stimulation frequencies were also not significantly differently between the two genotypes (fig. 6G-I). The paired-pulse ratios, either $\text{uIPSC}_2/\text{uIPSC}_1$ or $\text{uIPSC}_{\text{Last}}/\text{uIPSC}_1$ were not different between WT and

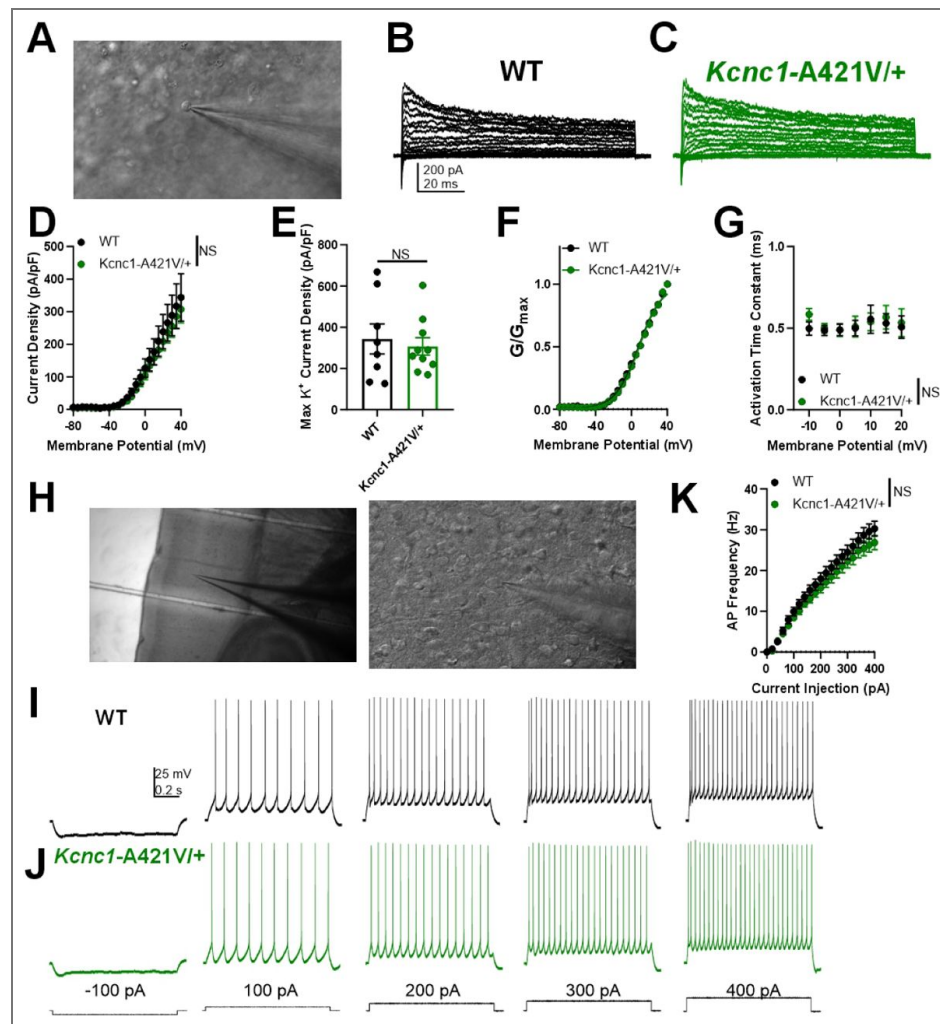


Figure 5. Unaltered potassium currents and physiological function of excitatory neurons in P16-21 *Kcnc1-A421V/+* mice.

A. Representative image of a neocortical excitatory cell being recorded in the outside-out nucleated macropatch configuration. **B-C.** Example family of traces of voltage-gated K^+ channel currents from a excitatory cell from WT (**B**, black) and *Kcnc1-A421V/+* (**C**, green) mice. **D.** Average voltage-gated K^+ channel current density for WT ($n=8$ macropatches, $N=3$ mice) and *Kcnc1-A421V/+* ($n=10$, $N=3$ mice) relative to membrane potential. **E.** Peak voltage-gated K^+ channel current density in WT and *Kcnc1-A421V/+* mice. **F.** Normalized voltage-dependent activation curves for WT and *Kcnc1-A421V/+* mice. **G.** Average voltage-dependent activation time constant for the voltage-gated K^+ channel currents. **H.** Representative images taken at 10X (left) and 40X (right) magnification of a layer IV neocortical excitatory cell recorded in the whole-cell configuration to characterize intrinsic excitability. **I-J.** Representative example traces showing excitatory cell AP generation in WT (**I**, black) and *Kcnc1-A421V/+* (**J**, green) in response to depolarizing current injections. **K.** Average relationship between excitatory cell AP frequency in response to a range of current injections for WT ($n=23$ cells, $N=3$ mice) and *Kcnc1-A421V/+* ($n=22$, $N=3$ mice). Data are shown as mean $\pm$ SEM and all results failed to reach significance determined via Repeated-measures Two-way ANOVA or unpaired t-test.

Kcnc1-A421V/+ (fig. 6J-K). Finally, we did not detect a significant difference in synaptic latency of the uIPSC between WT and *Kcnc1*-A421V/+ neuron pairs (fig. 6L). These data suggest that, despite expression of Kv3.1 in the axon and synaptic terminal in WT mice, neocortical PV-IN-mediated inhibitory synaptic neurotransmission remains intact in *Kcnc1*-A421V/+ mice at this juvenile developmental timepoint. Yet, inhibitory transmission will be impaired in *Kcnc1*-A421V/+ mice secondarily to the abnormal excitability and impaired spike generation of PV-INs.

In nine of 61 neuron (14.8%) pairs for WT mice and six of 43 pairs (14.0%) for *Kcnc1*-A421V/+ mice at P16-P21, we observed unitary excitatory synaptic currents (uEPSCs) from the excitatory neuron onto the PV-IN which allowed us to investigate whether there were abnormalities in excitatory neurotransmission (Figure 6—figure supplement 1A-D). The frequency-dependent rate of failure showed no significant effect for genotype (Figure 6—figure supplement 1E). Lastly, the magnitude (Figure 6—figure supplement 1F), paired-pulse ratio (Figure 6—figure supplement 1G), and the latency (Figure 6—figure supplement 1H) of the uEPSCs were not significantly different between WT and *Kcnc1*-A421V/+ neuron pairs indicating that excitatory neuron synaptic transmission onto PV-INs is not altered in juvenile *Kcnc1*-A421V/+ mice.

PV-IN synaptic neurotransmission is altered in adult *Kcnc1*-A421V/+ mice

As Kv3.1 exhibits developmentally-regulated expression, we next assessed PV-IN mediated inhibitory neurotransmission in young adult (P32-42) *Kcnc1*-A421V/+ mice relative to WT controls (fig. 7). As in the younger mice, we recorded pairs of cortical PV-IN and excitatory neurons that were synaptically connected—14 of 36 pairs (38.9%) in WT mice ($N=8$ mice) and 13 of 36 pairs (36.1%) in *Kcnc1*-A421V/+ mice ($N=8$ mice; fig. 7A-D). We did not detect differences in failure rates between WT and *Kcnc1*-A421V/+ mice over a range of stimulation frequencies (fig. 7E). Notably, the average magnitude of the first five uIPSCs was significantly increased in *Kcnc1*-A421V/+ relative to WT recordings (** $P<0.01$ at 20 Hz, * $P<0.05$ at 40 Hz, and * $P<0.05$ at 80 Hz; fig. 7F-H). Additionally, we observed a reduced paired-pulse ratio for the second uIPSC relative to the first uIPSC in *Kcnc1*-A421V/+ neuron pairs from young adult mice across a range of stimulation frequencies (* $P<0.05$; fig. 7I), but found no genotype effect in the average ratio between the last uIPSC to the first uIPSC (fig. 7J). The uIPSC latency measured from AP peak to onset of the synaptic event was not significantly different between WT and *Kcnc1*-A421V/+ neuron pairs (fig. 7K). Overall, these complex results align with prior work showing that Kv3.1 is an important regulator of synaptic neurotransmission in PV-INs and suggests that synaptic dysfunction may contribute to the pathogenesis of epilepsy in *Kcnc1*-A421V/+ mice in a developmentally determined manner.

2-Photon *in vivo* calcium imaging reveals paroxysmal hypersynchronous discharges and altered neuronal excitability in *Kcnc1*-A421V/+ mice

We then utilized two-photon (2P) calcium imaging to investigate neural activity in neocortical circuits in both WT and *Kcnc1*-A421V/+ mice (>P50) in layer II/III of primary somatosensory cortex *in vivo* (Figure 8A). We observed striking instances of paroxysmal hypersynchronous discharges in the neuropil signal of seven of seven *Kcnc1*-A421V/+ mice examined, but never in WT mice (Figure 8B-D). During each hypersynchronous discharge, the mouse was stationary and non-ambulatory, but displayed a brief diffuse twitch involving the facial musculature and bilateral limbs (Figure 8—video supplement 1). Taken together, these *in vivo* imaging results indicate that *Kcnc1*-A421V/+ mice exhibit paroxysmal synchronous discharges that may correlate with seizures, potentially myoclonic seizures.

Because the occurrence of large-amplitude hypersynchronous discharges in the neuropil contaminated the somatic signal in these recordings, we performed *in vivo* 2P imaging in a separate cohort of WT and *Kcnc1*-A421V/+ mice that were co-injected with a pan-neuronal soma-tagged GCaMP8m and an S5E2 enhancer-driven tdTomato to identify PV+ interneurons.

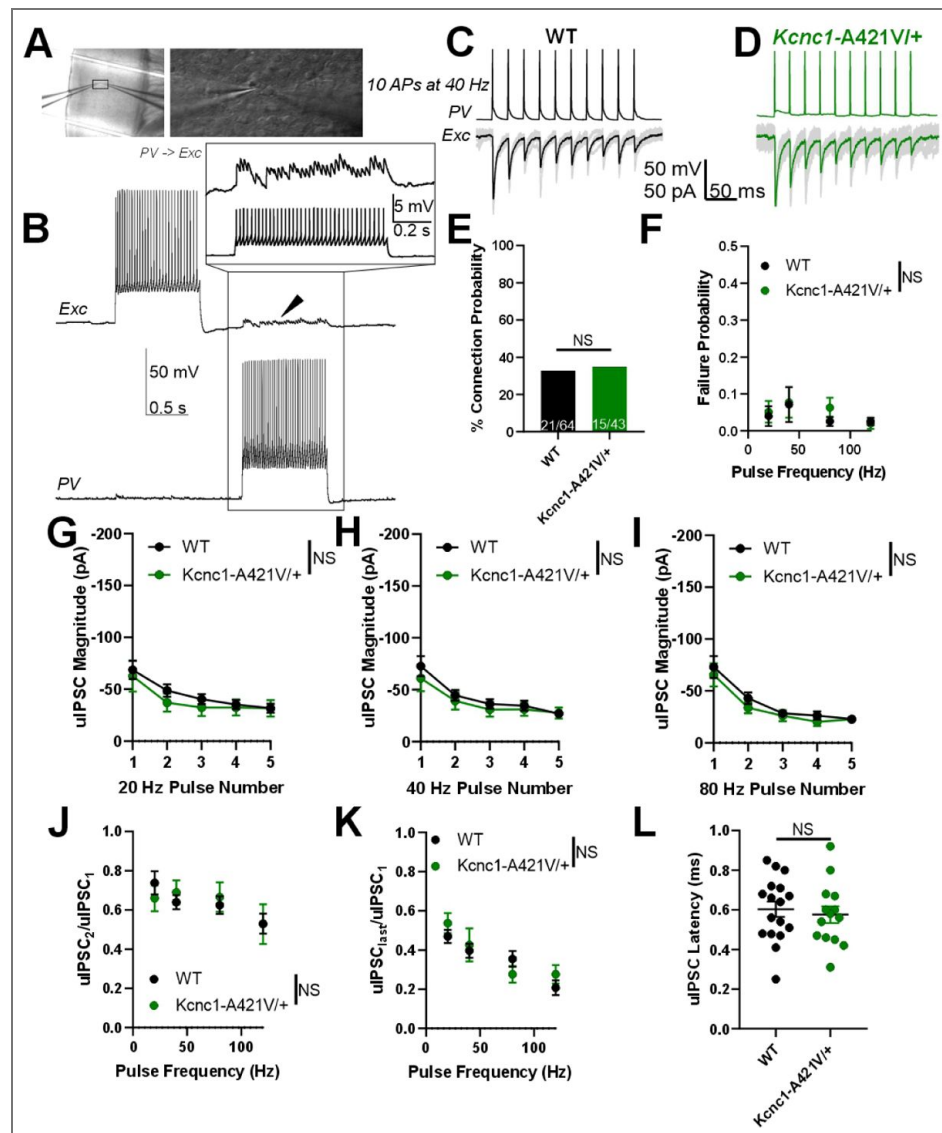


Figure 6. Juvenile (P16-21) *Kcnc1-A421V/+* mice exhibit normal PV-IN mediated inhibitory synaptic neurotransmission.

A. Representative images showing simultaneous whole-cell patch-clamp recordings of cortical PV-IN and nearby excitatory cell (left, 10x magnification; right 40x magnification). **B.** Example traces of a PV-IN and excitatory cell pair in which generation of APs in the PV-IN (bottom trace) is sufficient to induce clear unitary inhibitory postsynaptic potentials (uIPSPs) in the excitatory cell (arrowhead, top trace). The inset shows an example view of the individual uIPSPs corresponding to each AP in the PV-IN. **C-D.** Example traces of uIPSCs in both WT (**C**) and *Kcnc1-A421V/+* pairs of synaptically-connected neurons. 10 APs were generated in the PV-IN (top trace) at 40 Hz and the uIPSCs are shown below. The black and green traces are the averages of numerous individual sweeps shown in gray. **E.** Probability of synaptic connection between PV-IN and excitatory cell in WT ($n=21$ 64 pairs from $N=7$ mice) and *Kcnc1-A421V/+* ($n=5$ of 43 pairs from $N=11$ mice). **F.** Average failure probability relative to presynaptic stimulation frequency in WT and *Kcnc1-A421V/+* neuron pairs. **G-I.** Average uIPSC magnitude for the first five APs in WT and *Kcnc1-A421V/+* at 20 Hz (**G**), 40 Hz (**H**), and 80 Hz (**I**) stimulus frequencies. **J-K.** Paired-pulse ratios for both WT and *Kcnc1-A421V/+* mice relative to stimulus frequency. Ratio of second uIPSC to the first is provided in **J**, while **K** displays the average ratio of the last uIPSC to the first. **L.** Average latency from peak of presynaptic AP to peak of the uIPSC in WT and *Kcnc1-A421V/+* mice. Data are shown as mean $\pm$ SEM and all results failed to reach significance determined via repeated-measures Two-way ANOVA or unpaired t-test.

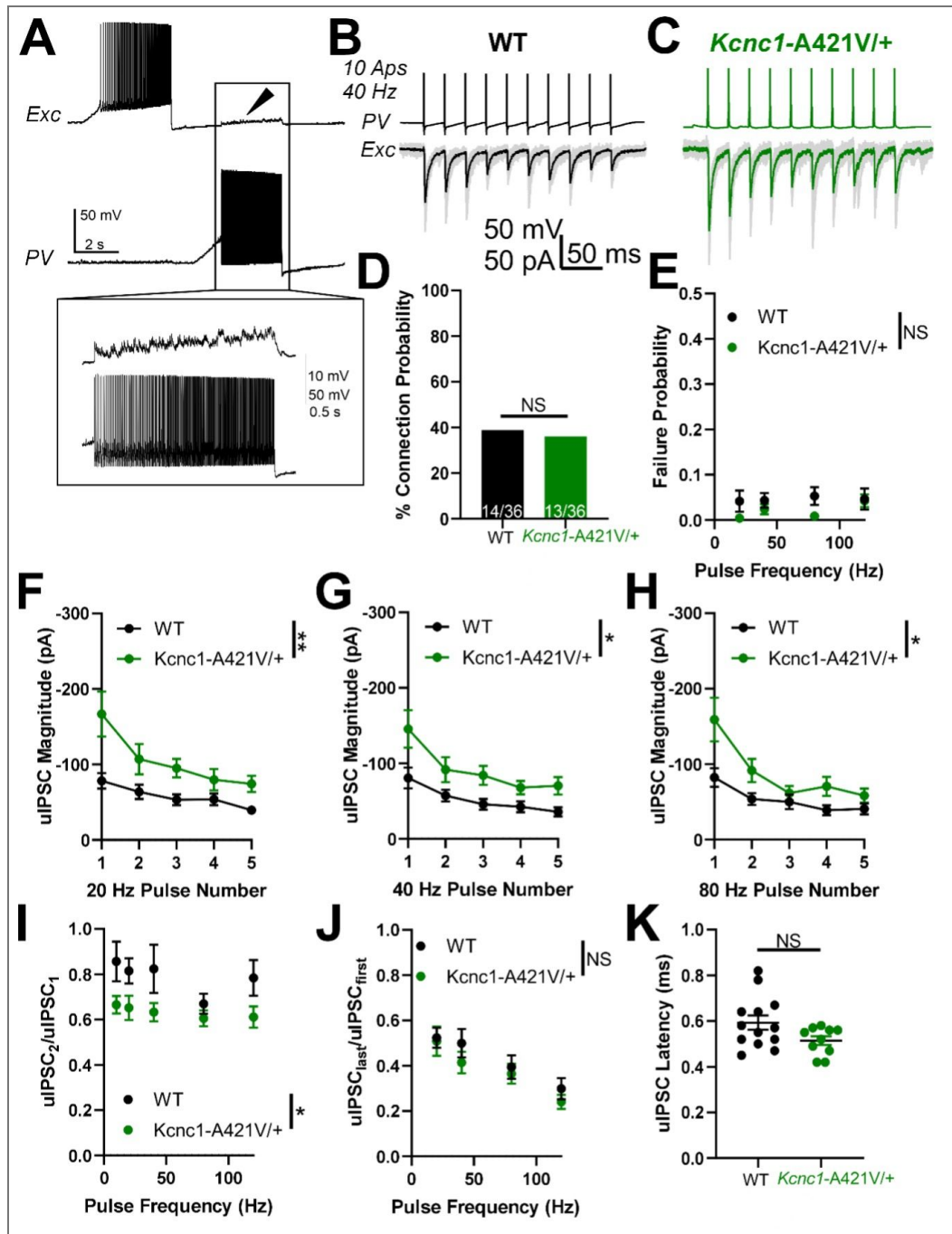


Figure 7. Adult (P32-42) *Kcnc1-A421V/+* mice exhibit altered PV-IN mediated synaptic neurotransmission.

A. Example presynaptic cortical PV-IN and post-synaptic excitatory neuron. Arrowhead and inset display the uIPSPs induced in the postsynaptic cell when the PV-IN generates APs. **B-C.** Example traces of uIPSCs in both adult WT (**B**) and *Kcnc1-A421V/+* (**C**) pairs of synaptically-connected neurons. 10 APs were generated in the PV-IN at 40 Hz and the evoked uIPSCs are displayed in the trace below where the black and green traces are the averages of numerous individual sweeps shown in gray. **D.** Connection probability between WT (14 of 36, $N=8$ mice) and *Kcnc1-A421V/+* (13 of 36, $N=8$ mice) pairs PV-INs and nearby excitatory cells. **E.** Average frequency-dependent rate of failure for the first five APs in adult WT and *Kcnc1-A421V/+* neuron pairs. **F-H.** Average uIPSC magnitude of the first five APs for adult WT and *Kcnc1-A421V/+* at 20 Hz (**F**), 40 Hz (**G**), and 80 Hz (**H**). **I-K.** Paired-pulse ratios for WT and *Kcnc1-A421V/+* neuron pairs (uIPSC₂/uIPSC₁ provided in **I**, uIPSC_{last}/uIPSC_{first} provided in **J**) relative to stimulus frequency. **K.** Average latency from AP peak to onset of the uIPSC in WT and *Kcnc1-A421V/+* mice. Data are shown as mean $\pm$ SEM or individual values and significance (* $P<0.05$, ** $P<0.01$) was determined using unpaired t-test or Repeated-measures Two-way ANOVA.

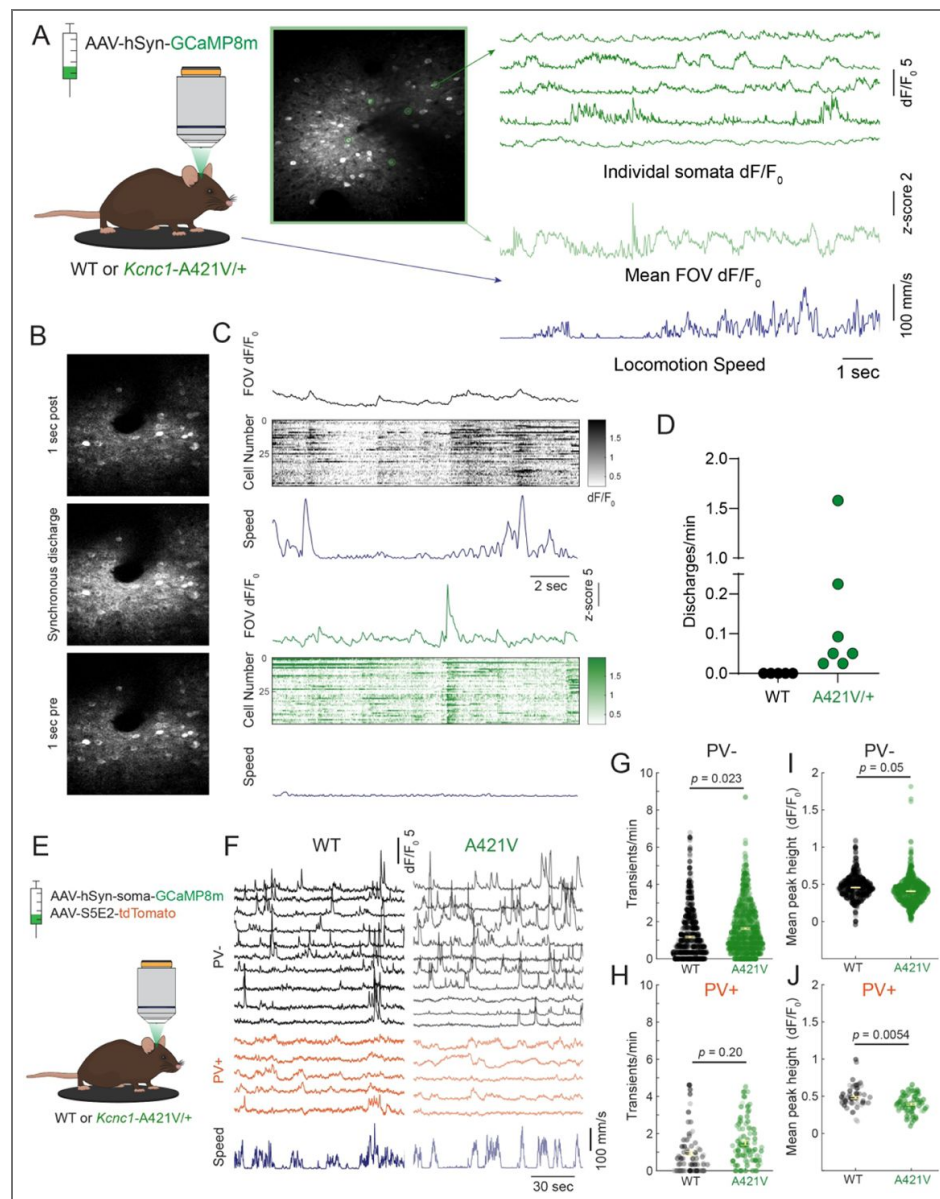


Figure 8. *In vivo* 2P calcium imaging reveals paroxysmal hypersynchronous discharges and altered neuronal excitability in *Kcnc1-A421V/+* mice.

A. Experimental setup for *in vivo* 2P calcium imaging with representative calcium transients from cells expressing AAV-hSyn-GCaMP8m and mean dF/F_0 of the whole field of view (FOV) aligned to locomotion speed. **B.** Representative 2P field of view during a hypersynchronous discharge in a *Kcnc1-A421V/+* mouse. **C.** Mean dF/F_0 of the field of view (top), calcium transients of individual somata (middle), and locomotion speed (bottom) during a paroxysmal discharge in the *Kcnc1-A421V/+* mouse relative to typical baseline activity shown in a WT mouse. Note that, in contrast to epochs of low-amplitude synchronization in WT associated with transition from quiet wakefulness to locomotion, there is no locomotion during the larger-amplitude hypersynchronous discharges identified in *Kcnc1-A421V/+* mice. **D.** Frequency of paroxysmal discharges in each mouse (N=5 WT, N=7 *Kcnc1-A421V/+* mice). **E.** Experimental design for *in vivo* 2P calcium imaging of somata positive (PV+) and negative (PV-) for parvalbumin. **F.** Example calcium transients of PV+ (bottom) and PV- (top) somata aligned to locomotion speed in a WT (left) and *Kcnc1-A421V* mouse (right). **G-H.** Transients per minute during quiet rest in (G) PV-(WT, N=885 cells, 4 mice, mean = 1.17; *Kcnc1-A421V*, n=1041 cells, N=3 mice, mean=1.63) and (H) PV+ cells (WT, N=4 mice, n=110 cells, mean=0.94; *Kcnc1-A421V*, N=3 mice, n=100 cells, mean=1.50). **I-J.** Mean peak height in (I) PV-(WT, mean=0.46; *Kcnc1-A421V*, mean=0.41) and (J) PV+ cells (WT, mean=0.48; *Kcnc1-A421V*, mean=0.40). Data points are shaded by mouse identity. Statistical comparisons were performed using mixed-effects modeling.

Interestingly, in this cohort, we did not detect hypersynchronous discharges in *Kcnc1*-A421V/+ mice, suggesting that this phenomenon is primarily localized to the neuropil. Analysis of PV- and PV+ cell activity revealed that, during epochs of quiet rest (when the mouse was stationary on the treadmill), PV-cells on average displayed more frequent calcium transients, whereas there was no change in the frequency of PV+ cell transients (fig. 8G-H). Of note, this held true even if only cells that displayed at least one transient throughout the recording were included in the analysis (Figure 8—figure supplement 1A-B). This effect is generally consistent with decreased perisomatic inhibition of excitatory cells in *Kcnc1*-A421V/+ mice. In line with this hypothesis, PV+ cells on average displayed lower amplitude transients, which could indicate that fewer APs underlie each calcium transient (Zhang et al., 2023). We also observed a decrease in the amplitude of the transients in both PV- and PV+ cells (fig. 8I-J). These decreases in the frequency of PV-cell transients and the height of PV+ cell transients were not seen during epochs where the mouse was running (Figure 8—figure supplement 1C-F). We also observed no differences in the overall percentage of cells that were active in *Kcnc1*-A421V/+ relative to WT mice (Figure 8—figure supplement 1G-H).

Spontaneous seizures and sudden death in *Kcnc1*-A421V/+ mice

To further investigate the nature of the events observed in *Kcnc1*-A421V/+ mice during *in vivo* 2P calcium-imaging, we performed continuous video electroencephalogram (EEG) monitoring for periods of two to seven days in (P24-48) WT and *Kcnc1*-A421V/+ mice (fig. 9). In eight of 12 *Kcnc1*-A421V/+ mice, we observed clear convulsive seizures including tonic, clonic, and tonic-clonic limb movements with loss of consciousness and fall, associated with an electrographic correlate (in the mice that exhibited seizures, the average seizure frequency was 0.62 ± 0.24 per day and event duration was 32.4 ± 15.7 seconds; fig. 9A-B; Figure 9—video supplement 1). We did not detect any seizures or other EEG abnormalities in four of four WT control mice (fig. 9C). As in our 2P *in vivo* calcium imaging experiments, we also observed brief diffuse jerks involving the face and limbs associated with large-amplitude spikes on the EEG (fig. 9A-C; Figure 9—video supplement 1). In eight of 12 mice, we observed brief runs of epileptiform spikes without apparent behavioral correlate (seizure duration was 14.5 ± 0.7 seconds; fig. 9C). Interestingly, we were also able to capture four *Kcnc1*-A421V/+ seizure-induced sudden death events on video-EEG with two additional occurrences with video only (Figure 9—video supplement 2). In each case, sudden death was directly preceded by a generalized tonic-clonic seizure with hindlimb extension, whereas nonfatal seizures did not lead to the hindlimb extension associated with the tonic phase of a generalized seizures (fig. 9A-B). Overall, our *in vivo* studies reveal that the *Kcnc1*-A421V/+ mouse recapitulates core features of *Kcnc1* DEE with a range of seizure types including myoclonic seizures.

Discussion

The recurrent pathogenic variant *Kcnc1* p.A421V leads to DEE characterized by treatment-resistant epilepsy with onset in the first year of life with multiple seizure types including myoclonic seizures, moderate to severe global developmental delay/intellectual disability, and variably present but mild nonprogressive ataxia. Further elucidation of the mechanistic links between *Kcnc1* variants, Kv3.1 subunit-containing K⁺ channel dysfunction, impairments in the intrinsic excitability of Kv3.1-expressing neurons, and synaptic and circuit neurophysiology is critical towards clarification of underlying disease pathomechanisms and development of potential therapeutic intervention. Our study reports the generation of a mouse model of *Kcnc1* DEE and determine the physiological mechanisms of disease at the level of ion channels, single neuron intrinsic excitability, and synaptic neurotransmission, as well as in circuits *in vivo*, within epilepsy-related brain regions.

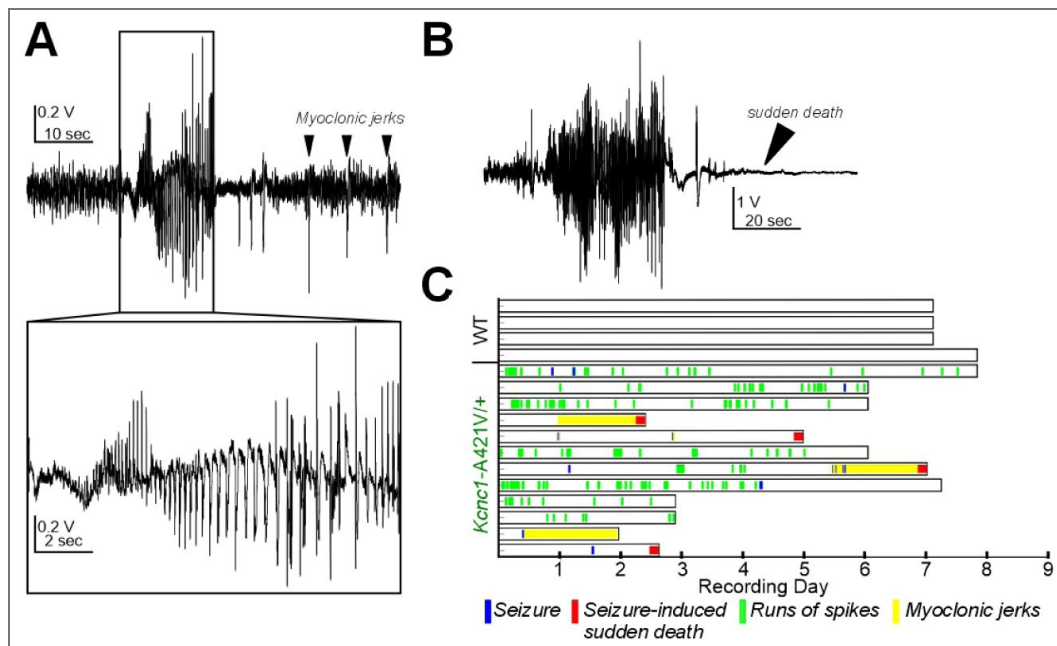


Figure 9. *Kcnc1-A421V/+* mice exhibit spontaneous seizures and seizure-induced death.

A. Representative example trace of the EEG collected from an adult *Kcnc1-A421V/+* mouse during a nonfatal seizure. After the seizure-related spike-wave discharges, there are large spikes that are associated with diffuse whole-body jerks. **B.** Representative generalized tonic-clonic seizure resulting in seizure-induced sudden death in a *Kcnc1-A421V/+* mouse. **C.** Raster plot indicating nonfatal seizures (blue bars), seizure-induced sudden death (red bars), interictal runs of spikes (green bars) without clear behavior manifestation, and periods of repetitive myoclonic seizures (yellow shading) for each mouse examined ($N=12$). Recordings in WT ($N=4$) control mice did not show epileptic seizures or runs of spikes.

Kv3.1 expression and function

Kv3.1 is specifically expressed in high-frequency firing neurons throughout the nervous system, including PV-INs in the neocortex, hippocampus, amygdala, and basal ganglia, as well as cells of the reticular thalamus, and Purkinje cells, granule cells, and molecular layer interneurons of the cerebellum (Chow et al., 1999 [↗](#)). Due to rapid activation and deactivation kinetics and unique voltage dependence (more positively shifted than any other K⁺ channel), Kv3.1 and other members of the Kv3 family (Kv3.2, Kv3.3) are associated with neurons that generate APs at particularly high frequencies >200 Hz (Erisir et al., 1999 [↗](#); Kaczmarek and Zhang, 2017 [↗](#)).

Kv3.1 knockout (Kv3.1^{-/-}) animals have previously been used to investigate the functional contribution of Kv3.1 to neuronal spiking. These mice have reduced body weight and altered sensory/motor function, as we observed in *Kcnc1*-A421V/+ mice, but do not display spontaneous seizures (Ho et al., 1997 [↗](#)). The identified impairments in intrinsic neuronal excitability of Kv3.1-expressing neurons were relatively subtle in Kv3.1 knockout mice: RTN neurons from Kv3.1^{-/-} mice exhibited slightly wider APs and a use-dependent spike broadening that produced a mild impairment in AP frequency; but, overall, there seemed to be functional and/or genetic compensatory upregulation of other Kv3 subfamily members in response to Kv3.1 deletion (Porcello et al., 2002 [↗](#)). Mice lacking Kv3.2 (Kv3.2^{-/-}), the other Kv3 family member highly expressed in PV-INs and which has near-identical biophysical properties, perhaps exhibit a somewhat more similar phenotype to that identified in the *Kcnc1*-A421V/+ mice, with impaired excitability of neocortical PV-INs and spontaneous seizures observed in a subset of mice (Lau et al., 2000 [↗](#)). Overall, for mechanistic reasons that are not yet completely clear, it seems that the heterozygous *Kcnc1*-A421V/+ mice reported here have a more severe phenotype than either Kv3.1 or Kv3.2 null mice. One possibility is that compensation shown to occur in knockout mice might not occur with heterozygous expression of a missense variant (i.e., the variant “escapes” compensation). Our data further supports the conclusion that the Kv3.1 A421V variant exerts a dominant-negative action on trafficking, as well as a possible additional effect on gating of Kv3.1/Kv3.2 heteromultimeric channels that do successfully traffic to the membrane. In contrast, Kv3.1 and Kv3.2 knockout mice influence only one Kv3 subfamily member (and drive compensatory upregulation of each other). Our results show that the A421V variant leads to decreased Kv3-like current in nucleated macropatches from neocortical PV-INs as well as impaired cell surface expression of Kv3.1 in neocortical PV-INs. Future studies should further clarify the precise mechanism whereby this missense variant impacts trafficking of and incorporation into heteromultimeric Kv3 channels in various subcellular compartments of PV-INs and other Kv3.1-expressing cells.

We found that Kv3.1-expressing neocortical PV-INs and cells of the RTN, but not excitatory cells (which do not express Kv3.1), were hypoexcitable in *Kcnc1*-A421V/+ relative to WT mice, generating fewer APs in response to depolarizing current injections. At P16-21, PV-INs exhibited a small but significant depolarization in resting membrane potential which may reflect delayed development of PV-INs in *Kcnc1*-A421V/+ mice (Goldberg et al., 2011 [↗](#)), be a direct result of altered potassium channel function, or could represent a compensatory response to intrinsic hypoexcitability. Beyond a role for PV-positive Kv3.1-expressing fast-spiking neurons, the extent to which specific cellular populations (e.g., cerebral cortex interneurons vs. neurons of the RTN) contributes to the overall epileptic/behavioral phenotype of the *Kcnc1*-A421V/+ mice remains unclear. Future studies using focal Cre injection or region-specific Cre drivers to express the A421V variant in a cell-type and/or region-specific restricted manner could be helpful for addressing these remaining questions.

Hypofunction of PV-INs has been associated with various types of epilepsy including, most notably, Dravet syndrome, a DEE driven by loss of function variants in *SCN1A* encoding the voltage-gated sodium channel subunit Nav1.1. Hence, Dravet Syndrome and *Kcnc1* DEE converge on specific impairment of cerebral cortex GABAergic inhibitory interneurons, and on PV-INs in particular. Yet, there are likely important differences between these syndromes which may explain the divergent clinical presentation in patients (Clatot et al., 2024 [↗](#)). For one, deficits in Kv3.1 in *Kcnc1* DEE may be differentially compensated by other Kv3 isoforms (perhaps remaining

uncompensated) when compared to a possible compensation for reduced Nav1.1 in Dravet syndrome by other voltage-gated sodium channel α subunits. Secondly, there is a differential cell type-specific expression pattern between Kv3.1 and Nav1.1 in the cerebral cortex, with Nav1.1 being also expressed in non-fast-spiking interneurons such as somatostatin and VIP-expressing interneurons (Tai et al., 2014 [↗](#); Rubinstein et al., 2015 [↗](#); Goff and Goldberg, 2019 [↗](#)), whereas Kv3.1 is largely specific for PV-INs. Yet, Kv3.1 is more prominently expressed in superficial layers of mouse neocortex with Kv3.2 more prominently expressed in deep layer PV-INs, while Nav1.1 appears to be expressed in PV-INs across layers of the neocortex. Our results here also indicate another point of divergence between mechanisms of Dravet Syndrome and *Kcnc1* DEE: in adult *Kcnc1*-A421V/+ mice, we observed an increase in the magnitude and altered paired-pulse ratio of PV-IN-mediated inhibitory postsynaptic currents relative to WT mice accompanied by no change in failure rate, which contrasts with the increased rate of failure and prolonged synaptic latency that we previously observed in PV-IN-mediated neurotransmission in Dravet syndrome (*Scn1a*^{+/-}) mice. We interpret the augmentation in postsynaptic current magnitude alongside reduced paired-pulse ratio observed in young adult (P32-42, after epilepsy onset) *Kcnc1*-A421V/+ mice to be generally consistent with a role for Kv3.1 in regulating neurotransmitter release by controlling spike-evoked calcium via presynaptic AP width, as shown previously (Goldberg et al., 2005 [↗](#)), although there could also be roles for secondary dysregulation of or compensatory alterations in other determinants of synaptic transmission (such as GABA receptor expression).

We did not find alterations in inhibitory synaptic neurotransmission at the P16-21 timepoint, despite the fact that PV-INs already exhibited markedly impaired intrinsic excitability and reduced magnitude of somatic voltage-gated K⁺ currents. These results may indicate that the physiological contribution of Kv3.1 in different subcellular regions (i.e. soma, dendrite, axon, terminal, etc.) and its corresponding role in regulating the associated physiological phenomena (AP generation, propagation, and neurotransmitter release) evolves over development. For example, the demonstrated impairment in trafficking of Kv3.1-A421V variant subunit containing Kv3 channels implies that distal synaptically-localized Kv3 channels may not contain variant subunits at early time points and hence local AP waveform at the synapse might remain largely normal via residual Kv3 channels containing WT Kv3.1 and/or Kv3.2 (or Kv3.3) subunits. A more detailed mechanistic explanation for this age-dependent effect would provide further insight into disease pathomechanisms and could explain why the epilepsy phenotype appears at/around the time of weaning and increases in severity in this mouse model (in stark contrast to *Scn1a*^{+/-} mice, where epilepsy severity decreases with age). However, this would require a detailed exploration of the specific composition of heterotetrameric Kv3 channels in WT vs. *Kcnc1*-A421V/+ mice in various subcellular compartments and across development, as suggested above.

In this study, we focused on PV-INs and excitatory neurons in somatosensory cortical layer II-IV, PV-INs of layer V, and PV-positive neurons of the reticular thalamus – epilepsy-linked brain regions – due to the prominent epilepsy phenotype and seizure-related early mortality observed in *Kcnc1*-A421V/+ mice. However, there are many other cellular populations across various brain regions that express Kv3.1 which could also be examined in future studies. As noted above, given that our mouse model expresses the knock-in A421V variant under the control of Cre recombinase, we are well-positioned to explore how cell type and developmental timing of altered Kv3.1 function might contribute to overall behavior phenotype.

***Kcnc1*-A421V/+ mice recapitulate the core phenotype of *Kcnc1* epilepsy seen in human patients**

Patients harboring the *Kcnc1*-p.A421V variant exhibit treatment-resistant epilepsy with various seizure types including myoclonic, focal, atypical absence, and generalized tonic-clonic seizures with onset in the first year of life (Cameron et al., 2019 [↗](#); Park et al., 2019 [↗](#)). The novel *Kcnc1*-A421V/+ mouse model well captured the range of seizure phenotypes observed: Spontaneous seizures with different behavioral manifestations were observed including myoclonic seizures, focal convulsive seizures, and generalized tonic-clonic seizures (those associated with hindlimb extension leading to sudden death). We directly observed abnormal neocortical neural activity in

the *Kcnc1*-A421V/+ mice in our *in vivo* 2P calcium-imaging experiments accompanied by behavioral correlates of myoclonic seizures, demonstrating this mouse represents a potentially useful model for study of mechanisms of spontaneous seizures – including myoclonic seizures – using 2P calcium imaging in awake, behaving mice. These experiments revealed that apparent myoclonic seizures were associated with hypersynchronous paroxysmal discharges seen across all neurons within the field of view, with prominent activation of the neuropil. Although we separately labeled fast-spiking PV-INs and other cells in our *in vivo* imaging experiments, we did not observe cell type-specific differences in recruitment of PV-INs and non-PV cells which might indicate a causal relationship between aberrant PV-IN activity and the hypersynchronous discharge. However, it is perhaps more likely that such seizures engaged diffuse brain networks, and the observed hypersynchronous discharges (and neuropil signal) were driven by distal activity. Nevertheless, these otherwise brief and intermittent events were clearly identified via 2P imaging which led to subsequent EEG studies confirming such events to be seizures. All *Kcnc1*-A421V/+ mice exhibited hypersynchronous discharges in our initial 2P experiments; yet, we did not observe such hypersynchronous discharges in all mice in which GCaMP expression was restricted to the soma. The basis of this apparent discrepancy is unclear but may support the conclusion that such events are generated distally and recruit the neurites of cells in the imaging field. Future studies should expand on the *in vivo* imaging and EEG completed here to more thoroughly investigate the cellular and network architecture of the neural activity underlying the spontaneously occurring myoclonic seizures. *Kcnc1*-A421V/+ mice may prove to be a particularly tractable model for the study of myoclonic seizures.

Beyond seizures, human patients harboring *KCNC1* variants show moderate to severe developmental delay and intellectual disability (Cameron et al., 2019 [DOI](#); Park et al., 2019 [DOI](#)). Young *Kcnc1*-A421V/+ mice showed developmental differences in body/brain weights; and although we did not detect other gross impairments in developmental milestones between postnatal days 5 and 15, which aligns with the expected developmental expression pattern of Kv3.1 and onset of fast-spiking around P15 (Goldberg et al., 2011 [DOI](#))(Okaty et al., 2009; Goldberg et al., 2011 [DOI](#)), adult (>P35) *Kcnc1*-A421V/+ mice exhibited cognitive impairment in both the Y-maze and Barnes maze test. These early developmental tests may have limited sensitivity to detect early subtle differences, and future studies should expand on this work with additional testing of cognitive, motor, social, and other behaviors.

The A421V *Kcnc1* variant leads to a loss of voltage-gated potassium channel function in PV-INs

Previous studies have reported that A421V is a loss-of-function variant when examined in *Xenopus* oocytes (Cameron et al., 2019 [DOI](#); Park et al., 2019 [DOI](#)). However, such work is conflicting as to the exact mechanism, with one paper showing evidence for a dominant-negative effect and another paper finding no evidence for dominant-negative action of the A421V variant. Differences in results obtained in *Xenopus* oocytes vs. mammalian cells may relate to culture conditions such as temperature, which is known to affect protein folding and trafficking. Our outside-out nucleated macropatch recordings of somatic voltage-gated K⁺ currents in brain slice showed clear approximately 50% reduction in K⁺ current density in PV-INs (but not excitatory cells) without changes in the biophysical properties of gating. In our examination of surface Kv3.1, we found a reduction in the amount of Kv3.1 that reaches the membrane in PV-INs from *Kcnc1*-A421V/+ mice. While we cannot rule out the possibility that some Kv3 tetramers at the cell surface contain Kv3.1-A421V subunits and act to decrease channel conductance (as suggested by our HEK cell recordings), our data is consistent with the view that the variant acts at least in part via incorporation of Kv3.1-A421V subunits into heterotetrameric Kv3 channels (likely in the endoplasmic reticulum) with impaired trafficking to the membrane. Consistent with this, a previous study identified A421V as exerting only slight steric hindrance relative to other developmental encephalopathy-causing *KCNC1* variants, supporting the conclusion that the profound impact of this variant on recorded currents is likely due mainly to impaired trafficking, potentially with some contribution via an impact on gating (Li et al., 2021 [DOI](#)). Similar structural

approaches may also help better determine the mechanism of trafficking deficiency and the extent to which A421V channels impair the trafficking of heteromultimeric Kv3 channels containing WT Kv3.1 and/or Kv3.2 subunits.

Kv3.1 as a drug target in epilepsy

Given the powerful influence of Kv3 channels on the excitability of neocortical PV-INs and neurons of the cerebellum, pharmacological modulators of Kv3.1 have been proposed as potential treatment for a range of neurological and psychiatric conditions including in a mechanistically targeted fashion for patients with *Kcnc1*-related disorders such as EPM7 (Rosato-Siri et al., 2015 [↗](#); Brown et al., 2016 [↗](#); Boddum et al., 2017 [↗](#); Chambers et al., 2017 [↗](#); Munch et al., 2018 [↗](#); Feng et al., 2024 [↗](#)). Previous reports showed that AUT-1 and related compounds facilitate greater firing frequency and spiking reliability of fast-spiking cells (Rosato-Siri et al., 2015 [↗](#); Brown et al., 2016 [↗](#); Boddum et al., 2017 [↗](#); Chambers et al., 2017 [↗](#); Munch et al., 2018 [↗](#); Feng et al., 2024 [↗](#)). Our study did not investigate the impact of Kv3.1 modulators in *Kcnc1*-A421V/+ mice. Yet, considering the decreased cell surface expression of Kv3.1 in PV-INs from *Kcnc1*-A421V/+ mice, one might predict limited efficacy of a small-molecule channel activator, unless such compounds could exert therapeutic effect via action on WT Kv3 channels not containing variant Kv3.1-A421V subunits. On the other hand, genetic approaches to either knock down expression of the A421V variant (perhaps using an antisense oligonucleotide) or boost expression levels of the WT Kv3.1, could be explored.

Limitations of the study

We provide evidence for a strong loss of total potassium current density and deficits in excitability in *Kcnc1*-A421V/+ PV-INs relative to WT, with the most severe alterations to excitability observed for PV-INs in superficial neocortical layers likely driven by a high relative expression of Kv3.1 vs. Kv3.2 in these cells (Chow et al., 1999 [↗](#)). While we also provided immunohistochemical evidence that variant Kv3.1 leads to impaired membrane trafficking of Kv3.1, the molecular details underlying how the variant induces an overall loss of potassium channel function remain to be definitively determined. For example, it is unknown what relative proportion of A421V-containing heterotetramers reach the cell surface, and, for any channels that do, it is yet unclear the extent to which such channels functionally gate and flux potassium current. Considering that the *Kcnc1*-A421V/+ mouse is significantly more severely affected in cellular and behavioral phenotype than Kv3.1 knockout mice, and that layer V PV-INs exhibit less severe impairment than layer II-IV PV-INs, we suspect that Kv3.1 A421V variant subunits exert a dominant negative influence on Kv3 channel expression and function; i.e., such variants impact all PV-IN Kv3.1/Kv3.2 heteromultimeric channels containing one or more Kv3.1 A421V variant subunits. This could be compounded by potential electrophysiological dysfunction of Kv3 channels containing Kv3.1 A421V variants that do traffic to the membrane. We focused our study on the global impact of the *Kcnc1*-A421V variant on mouse development, epilepsy, and neuronal physiology of selected neuron types in epilepsy-linked brain regions, using ActB-Cre to drive global expression from the blastocyst stage so as to best model the human condition. Future work using cell type-specific Cre-drivers or Cre delivery to restricted brain regions will enable greater mechanistic clarity in linking cell type and brain region to specific aspects of the mouse phenotype, including epilepsy and non-epilepsy comorbidities of cognitive and motor impairment. However, given the early onset of neurological dysfunction in our mice, specific expression of the variant using, for example, PV-Cre mice, might not yield greater mechanistic insight, as PV itself is not expressed at appreciable levels until at/beyond P10 in mice and hence efficient recombination and expression of the *Kcnc1*-p.A421V variant in PV-INs will likely not faithfully reproduce the appropriate developmental expression pattern.

Conclusion

In summary, we report a mouse model that recapitulates the core features of *Kcnc1* developmental and epileptic encephalopathy due to the recurrent K⁺ channel variant *Kcnc1*-p.A421V. *Kcnc1*-A421V/+ mice exhibit epilepsy with multiple seizure types including myoclonic seizures as well as cognitive impairment. This was associated with a pattern of specific impairments in intrinsic excitability and synaptic transmission consistent with Kv3 dysfunction, observed in Kv3.1-expressing neurons linked to epilepsy including neocortical PV-INs and neurons of the reticular thalamic nucleus, but not excitatory cells. Future studies and ongoing therapeutic development promise to expand this mechanistic understanding in pursuit of improved outcomes for patients with this severe yet currently incurable and untreatable disorder.

Methods and materials

Experimental animals

All procedures were approved by the Animal Care and Use Committee at the Children's Hospital of Philadelphia and were conducted in accordance with the published ethical guidelines by the National Institutes of Health. Male and female mice were used in approximately equal numbers in each experiment throughout the study and, while our study was not powered to detect sex differences *a priori*, we observed no significant interaction between sex and genotype in our study. In experiments on pre-weanling mice ages postnatal day 16-21, mice were housed with the dam; in post-weaning experiments, mice were group-housed by sex (≤ 5 mice/cage) and had access to food/water *ad libitum* in a temperature and humidity-controlled room with a 12:12 hour light:dark cycle.

Kcnc1-Flox(A421V)/+ mice were generated via a gene targeting strategy. The targeting vector contained part of intron 1 and exons 2-4 of the *Kcnc1* coding sequence flanked by LoxP sites. A point mutation C>T (Ala421Val) was introduced into exon 2 in the 3' homology arm. The linearized vector was electroporated into ES cells and homologous recombination was confirmed by Southern blot. The targeted ES cell clone was then injected into mouse blastocysts and founder animals were identified by coat color. Germline transmission was confirmed by breeding with C57BL/6 mice and subsequent genotyping of the offspring. Hence, in the absence of Cre recombinase, the inserted wild-type sequence is expressed; in the presence of Cre recombinase, there is Cre-mediated excision of the inserted wild-type sequence, leading to the expression of the modified *Kcnc1* allele. Genotyping was performed using primers designed to detect upstream and downstream LoxP sites. The presence of the knock-in point mutation site was confirmed via Sanger sequencing. The homozygous floxed *Kcnc1* males were crossed to hemizygous Pvalb-tdT BAC transgenic reporter females (JAX# 027395) for generation of double transgenic *Kcnc1*-Flox(A421V)/+:Pvalb-tdT/+ mice. The female offspring were then crossed to homozygous ActB-Cre males (JAX# 003376) for generation of experimental mice. This strategy produced roughly equal numbers of WT:ActB-Cre:Pvalb-tdT (WT) and *Kcnc1*-A421V/+:ActB-Cre:Pvalb-tdT triple transgenic mice of both sexes in which PV-INs were fluorescently labeled, all on a 50:50 C57BL/6N:6J genetic background. After generation and establishment of the line, routine genotyping was completed through Transnetyx automated genotyping services.

Immunohistochemistry

Mice were deeply anesthetized with isoflurane and underwent transcardial perfusion with 10 mL of 4% paraformaldehyde (PFA) in PBS. Whole brains were extracted and postfixed for 24 hours. Parasagittal brain sections were cut at 40 μ m intervals using a vibratome (Leica VT1200S). After washing in PBS, the slices were incubated in blocking solution (3% normal goat serum (NGS), 2% bovine serum albumin (BSA), 0.3% Triton X-100 in PBS) for 1 h at room temperature and then washed in PBS. The sections were then transferred into primary antibody solution containing rabbit Kv3.1b antibody (1:500; Alomone Labs APC-014) and mouse IgG1 parvalbumin antibody (EMD Millipore MAB1572) in blocking solution (1% NGS, 0.2% BSA, 0.3% Triton X-100 in PBS) and incubated at 4°C overnight. Next, sections were washed with PBS and incubated for 1h at room

temperature in secondary antibody solution containing goat anti-rabbit Alexa Fluor 488 (1:1000, Molecular Probes A11034) and goat anti-mouse IgG1 Alexa Fluor 568 (1:1000, Molecular Probes A21124) in blocking solution (2% BSA, 0.3% Triton X-100 in PBS). Finally, sections were washed 3 times with PBS, incubated with DAPI (1:1000, Thermo Fisher D3571) for 10 min, and washed again with PBS. Sections were mounted on glass slides using polyvinyl alcohol mounting medium with DABCO (Sigma 10981). Images were acquired from somatosensory cortex using a confocal microscope (Leica SP8) equipped with 10X and 40X objectives and image processing was performed with ImageJ software (NIH, USA).

Image Analysis

Parvalbumin-positive cells from somatosensory cortex layers II-IV were imaged and manually traced using the parvalbumin signal to define the cell soma and the DAPI signal to define the cell nucleus. A membrane compartment was defined as the outermost 1 micron of the parvalbumin-defined cell soma. A cytosol compartment was defined as the region inside the membrane compartment and outside the DAPI-defined nucleus. The mean Kv3.1b signal in the membrane compartment of each cell was measured and compared to the mean signal in the cytosolic compartment. To analyze cell density, parvalbumin-positive cells in somatosensory cortex layers 2-4 were imaged using a 10X objective and identified using automatic thresholding based on the ImageJ Triangle threshold method. Automatically identified cells were manually verified for quality control. All analysis was done blinded to genotype, and statistical comparisons were made using an unpaired t-test.

Developmental Milestone Assessments

WT and *Kcnc1*-A421V/+ mice were examined for developmental milestones from postnatal day 5 through 15 as previously described (Armstrong et al., 2020 [↗](#); Feng et al., 2024 [↗](#)). Mice were examined manually for onset of fur appearance, eye opening, ear canal opening, incisor eruption, and elevation of the head and shoulders as previously described. Auditory startle was determined by examining for flinching in response to presentation of a loud stimulus (hand clap near cage). Horizontal and vertical screen tests, cliff avoidance, quadruped walking, and negative geotaxis were completed as previously described (Armstrong et al., 2020 [↗](#); Feng et al., 2024 [↗](#)). Statistical comparison of age-dependent brain and body weights were determined using a Two-way ANOVA.

Behavioral Tasks

Kcnc1-A421V/+ mice and wild-type littermates aged P35-65 were used for behavioral analysis. Animals were acclimated to the experimental room for 1 hour before behavioral testing. Mice were video tracked using an ANY-Maze system (Stoelting).

The Barnes maze consisted of a circular platform 90 cm in diameter, which was elevated 1 m above the floor. The periphery of the platform was equipped with 20 evenly spaced holes, 5 cm in diameter. The platform could rotate freely and had a secondary black acrylic portion beneath it, which was used to block the bottom of 19 of the holes and hold the escape chamber beneath the 20th hole. The platform was illuminated by an overhead UFO LED flood light at 1300 lux. For navigation, visual cues are placed in the walls surrounding the platform at eye level (just above the table height). Before the first trial of the acquisition phase, each mouse was gently placed in front of the escape hole and allowed to climb down to the escape chamber. The mouse was left inside the escape box for 3 min to get familiarized with the procedure and reduce anxiety levels. The first trial of the acquisition training phase started immediately after this habituation phase. At the beginning of each trial, mice were placed inside an opaque, open-topped container located in the center of the maze. After 10 s, the container was removed, and the trial started. Each trial ended when the mouse entered the escape compartment, or after 150 s had passed, whichever occurred first. Mice underwent two acquisition trials per day over 4 days, with an intertrial interval of 30 min. The latency of the mouse to enter the escape hole was recorded for each trial. Mice that did not enter the escape compartment were assigned a latency of 150 s. 72 hours after

the final trial of acquisition training, mice underwent a 180 s probe trial in which the escape box was removed from the Barnes maze apparatus. The time spent in the quadrant that previously had the escape box was measured to account for long-term spatial memory.

Spontaneous alternation performance was assessed in a symmetrical Y-maze under reduced light conditions (approximately 30 lux). The maze consisted of three arms, each 50 cm long, positioned with a 120-degree angle between them. Mice were allowed to explore the maze for 8 minutes. The percentage of spontaneous alterations was calculated as the number of alternations (consecutive entries into three different arms) divided by the total possible alternations (total arm entries minus 2) and multiplied by 100.

In vivo two-photon calcium imaging

For stereotaxic viral injection and cranial window implantation, WT and *Kcnc1*-A421V/+ mice age >P35 were anesthetized with isoflurane (induction at 3–4%; maintenance at 1–1.5%). A 3 mm craniotomy over primary somatosensory cortex (centered at 1 mm posterior, 3 mm lateral to the bregma) was made, and a Nanoject III (Drummond Scientific) was used to inject either 60 nL of AAV9-syn-jGCaMP8m-WPRE (Addgene #162375) alone or mixed with PHP.eB-E6-S5E2-dTom-nlsdTom (Addgene #135630). Each virus was diluted to a titer of 2×10^{12} in sterile PBS. Injections were delivered at a depth of 300–500 μm and a rate of 1 nL/sec using a 50 μm diameter beveled tip glass pipette. A 3 mm circular coverslip glued to a 5 mm circular coverslip was affixed over the craniotomy and a custom stainless steel headbar was cemented to the skull. Mice were given buprenorphine-SR 0.5 mg/kg, cefazolin 500 mg/kg, and dexamethasone 5 mg/kg perioperatively and monitored for recovery from surgery and development of any signs of infection for 48 h following surgery. 2–3 weeks following the cranial window implantation, mice were habituated to head fixation on the Mobile HomeCage Apparatus (Neurotar) in a custom chamber with transparent siding for 2 days or until the mouse showed spontaneous running bouts and the absence of escape or freezing behaviors. Airflow into the Mobile HomeCage stage provided ~40–45 dB pink noise during the experiment. Locomotion speed was tracked by the Mobile HomeCage locomotion tracking software (Neurotar). An infrared (IR) 850 nm light source and IR CCD camera (Grasshopper 3, Teledyne FLIR) was used to record mouse behavior during imaging. 2P imaging was performed on an Ultima 2P-plus microscope (Bruker) equipped with a resonant scanner using a tunable femtosecond-pulsed IR InSight X3 laser (Spectra-Physics) with output controlled by a Pockels cell (Conoptics). Imaging was performed at a sampling rate of 30 Hz using an excitation wavelength of 950 nm through a 16X/0.8 NA water immersion objective (Nikon) with an additional 2X optical zoom. GCaMP8m and dTomato signal were collected with a gallium arsenide phosphide photodetector (H742240, Hamamatsu) and a multi-alkali detector (R3896, Hamamatsu) respectively. 10-minute recordings were performed across 3–5 distinct fields of view in each mouse.

In vivo 2-photon calcium imaging analysis

Cell detection and extraction of neuronal activity were performed as previously described (Goff et al., 2023 [↗](#)). Briefly, we used the Suite2p package to perform a nonrigid registration, detection of cell ROIs, and extraction of fluorescence values from ROIs (Pachitariu et al., 2017 [↗](#)). Manual quality control was performed on all potential ROIs by a blinded experimenter. Fluorescence values extracted from each ROI and the mean fluorescence of the whole FOV were transformed into $\frac{dF}{F_0}$ values, which was calculated as $\frac{dF}{F_0} = \frac{F(t) - F_0}{F_0}$, where F is the 10th percentile of the fluorescence trace, adjusted by using a linear interpolation between the average F_0 values for each 1000 frames of the recording. Paroxysmal synchronous discharges were detected in the traces of the mean dF/F_0 for the whole field of view by highpass filtering the trace at 1 Hz, then identifying peaks with a zscore greater than 5. Transients were identified by first smoothing the data using a moving mean filter, calculating the zscore of the trace, then using the findpeaks function to find transients with a minimum zscore of 1, and a minimum distance of 200 ms between individual transients, and a minimum prominence of a zscore of 0.5. Analyses were repeated with different threshold values for peak finding to ensure robustness.

In vivo Seizure Monitoring

Chronic video-EEG recordings were collected as previously described (Feng et al., 2024²) using a wireless EEG system (Data Sciences International, St. Paul, MN) and standard surgical approaches. Mice aged P24-48 were anesthetized with isoflurane and four small burr holes were made in the skull above the mouse motor and barrel cortices. A telemetry device containing four electrode leads was implanted subcutaneously on the back of the mouse. The insulation on the positive and negative leads was removed and the exposed wire was manually bent to create a relatively flat terminal to place on the surface of the dura. The leads were stably secured in the head cap via adhesive cement (C&B-Metabond, Parkell Inc, Brentwood, NY). Once the incision was sutured, the mice were given local treatment with antibiotic ointment (OTC Generics, Patterson Veterinary, Houston, TX) and were singly housed in home-cages for recovery. Continuous video and EEG recordings were collected for two to seven days using the Ponemah Software System (DSI, St. Paul, MN) and the EEG signal was acquired at 500 Hz. The aligned video and EEG signals are accessed using Neuroscore (DSI) software. First, the EEG signals are preprocessed by filtering with a powerline filter (60 Hz notch filter) followed by 1Hz high pass filtering. Then, an analysis protocol is designed in the Neuroscore software for spike and spike train detection to determine periods of abnormal EEG. Spikes were detected when the EEG signal exhibited an amplitude of 200 μ V and was greater than the root-mean-squared value of the activity within the preceding minute. A spike train was detected when at least five spikes were detected, the inter-spike interval was between 0.05 and 0.6 seconds, and the duration of the train was at least three seconds. Detected spikes and spike trains were then analyzed manually alongside the recorded video for behavior to classify events into runs of epileptiform spikes, spontaneous seizures, myoclonic seizures, or seizure-induced sudden death events. Runs of epileptiform spikes were defined as detected spike-trains that lacked clear behavioral manifestation. Spontaneous seizures were detected as spike trains of at least three seconds coincident with mouse loss of balance and/or convulsive movements. Myoclonic seizures were identified by large and brief (<200 ms) single epileptiform spikes that were coincident with periodic whole-body spasm-like movements. All seizure-induced sudden death events were characterized as spontaneous seizures which culminated in hindlimb extension and suppressed EEG signal. All EEG data along with identified events were exported into MATLAB to produce raster plots.

Plasmid preparation, cell culture and transfection

Recordings of Kv3.1 in HEK cells were conducted as previously described (Clatot et al., 2023³). Briefly, a cDNA plasmid encoding human KCNC1 (reference sequence NM_001112741.2) and A421V variant were synthesized and subcloned into a pCAG plasmid. HEK-293T cells (ATCC, CRL-3216) were grown at 37°C and 5% CO₂ DMEM supplemented with 10% heat-inactivated fetal calf serum and 1% penicillin-streptomycin in 35 mm dishes. Cells were transfected with 0.1 μ g of pCAG.EGFP and 0.2 μ g of either wild-type (WT) or variant hKCNC1 cDNA using PolyFect transfection reagent (QIAGEN; Germantown, MD, U.S.A.) as instructed by the manufacturer. After 24 hours, cells were treated with trypsin and seeded at low-density and single GFP-positive cells were identified for patch-clamp experiments.

HEK Cell Patch-clamp Electrophysiology

Whole-cell patch-clamp electrophysiology experiments were performed at room temperature using an Axopatch 200B amplifier (Molecular Devices, Sunnyvale, CA) in an extracellular Tyrode's solution consisting of the following: 150 mM NaCl, 2 mM KCl, 1.5 mM CaCl₂, 2 mM MgCl₂, 10 mM HEPES and 10 mM glucose; pH was adjusted to 7.4 with NaOH. Intracellular solution contained, in mM: 125 KCl, 25 KOH; 1 CaCl₂, 2 MgCl₂, 4 Na₂-ATP, 10 EGTA, 10 HEPES, with pH adjusted to 7.2 with KOH and osmolality to 305 mOsm/L with sucrose.

Patch pipettes were fashioned from thin-walled borosilicate glass (Harvard apparatus, Holliston, MA, U.S.A.) and fire-polished (Zeitz) to a final resistance of 1.7–2.5 Mohm in the whole-cell recording configuration. Voltage errors were reduced via series resistance compensation. Currents were filtered at 2 kHz by a low-pass Bessel filter and digitized at 30 kHz. Data were acquired with

pClamp 11 and analyzed with Clampfit (Axon Instruments, San Jose, CA, U.S.A.). Transient potassium currents were measured by performing 100-ms step depolarizations to between -85 and +55 mV in increments of 5 mV from a holding potential of -120 mV. Activation conductance was normalized, plotted against voltage, and fit with a Boltzmann function. All recordings and analysis were performed blinded to experimental group.

HEK Cell Data Analysis

Recordings were obtained from at least $n = 10$ cells from multiple transfections. Data were analyzed using custom-written MATLAB scripts, Clampfit 11, and Sigma Plot 11 (Systat Software, Inc., San Jose, CA, U.S.A.). Results are presented as the mean $\pm$ standard error of the mean (SEM) and significance was determined using a one-way ANOVA.

Patch-clamp Electrophysiology Recordings in Acute Brain Slices

Acute brain slices were prepared as previously described (Wengert et al., 2021 [DOI](#), 2019 [DOI](#), 2022 [DOI](#); Kaneko et al., 2022 [DOI](#)). Mice were deeply anesthetized with 5% isoflurane and decapitated so that the brain could be quickly removed and chilled in ice-cold artificial cerebrospinal fluid (ACSF) containing in mM: 125 NaCl, 2.5 KCl, 1.25 NaH_2PO_4 , 2 CaCl_2 , 1 MgCl_2 , 2 Na-pyruvate, 0.5 L-ascorbic acid, 10 D-glucose, and 25 NaHCO_3 (osmolarity = 305 mOsm). Horizontal slices (300 μm thick) were gently collected, bisected, and transferred to 37°C ACSF for 30 minutes. The slices were then kept at room temperature for up to 5 hours. Throughout all procedures the ACSF was constantly bubbled with 95/5 O_2/CO_2 carbogen gas. For recording, acute brain slices were gently transferred to a recording chamber on the stage of an upright light microscope (Olympus) and were superfused at 3 mL/min (SmoothFlow Model Q100-TT-ULP-ES TACMINA Corporation, Japan) with ACSF warmed to 32°C (TC-324C Warner Instruments).

Recordings of intrinsic excitability and synaptic neurotransmission

Cortical PV-INs in layers II-IV of the somatosensory cortex were targeted for patch-clamp electrophysiology recordings due to their red fluorescence and non-pyramidal morphology while excitatory cells (spiny stellate and pyramidal neurons) were reliably targeted based on cellular morphology and orientation and the identity of cells was confirmed by examining electrophysiological properties (i.e. fast-spiking for PV-INs, and regular spiking for excitatory cells). Red-fluorescent neurons were also easily identifiable in the reticular nucleus of the thalamus. Whole-cell patch-clamp electrophysiology recordings were collected using borosilicate patch pipettes (Sutter Instruments OD=1.5 mm; ID=0.86 mm) pulled using either a P-97 or P-1000 Flaming-Brown micropipette puller (Sutter Instruments) to have resistance values of 2.5-4.5 M Ω when filled and placed in the bath solution.

Voltage-gated K^+ channel function was assessed through recordings of outside-out nucleated macropatches as done previously (Bekkers, 2000 [DOI](#); Korngreen and Sakmann, 2000 [DOI](#)). After achieving the whole-cell recording configuration in PV-INs, light negative pressure was applied to bring the nucleus to the pipette. Slow retraction of the pipette enabled a large piece of the somatic membrane to be pulled off while reestablishing the gigaseal. Upon formation of the macropatch, the membrane capacitance was offset and the series resistance was confirmed to be below 12 M Ω . Voltage-gated K^+ channel currents were generated by 100-ms voltage steps from -80 to 40 mV in increments of 5 mV. The macropatch recordings used an internal solution containing in mM: 130 K-gluconate, 6.3 KCl, 1 MgCl_2 , 10 HEPES, 0.5 EGTA, 10 phosphocreatine-Tris, 4-ATP (magnesium salt), 0.3-GTP (disodium salt) and was adjusted with KOH to have a final pH of 7.3

The pipette internal solution for recordings of intrinsic excitability and synaptic neurotransmission contained in mM: 65 K-gluconate, 65 KCl, 2 MgCl_2 , 10 HEPES, 0.5 EGTA, 10 phosphocreatine-Tris, 4-ATP (magnesium salt), 0.3-GTP (disodium salt) and was adjusted with KOH to have a final pH of 7.3 and an osmolarity of 290 mOsm. This internal had a calculated reversal potential for chloride at $E_{\text{Cl}} = -17$ mV to enable larger signal to noise ratio for recordings of unitary inhibitory post-synaptic currents (uIPSCs). Membrane potential was sampled at 100 or 33 kHz with a Multiclamp 700B amplifier (Molecular Devices), filtered at 5 kHz or in some cases 2 kHz, and

digitized using a Digidata 1550B digitizer (Molecular Devices), and acquired using the pClamp 11 software suite (Molecular Devices). Cells were discarded if the resting membrane potential was visibly unstable or more depolarized than -50 mV, or if the access resistance changed by >20% over the duration of the experiment. Throughout the study, we left our liquid-junction potentials uncorrected.

Intrinsic excitability of individual neurons was assessed similarly to previous reports (Wengert et al., 2019 [DOI](#), 2021 [DOI](#)). For AP properties, analysis was completed on the first AP generated in response to a ramp of depolarizing current (100 pA/sec). Resting membrane potential and/or spontaneous excitability were determined as the median value of a Gapfree recording collected 2 minutes after achieving the whole-cell configuration. Upstroke and downstroke velocity were determined as the maximum and minimum of the first-derivative of the first AP. Threshold was determined as the membrane potential in which the first-derivative exceeded 5% of the upstroke velocity. Amplitude was calculated as the difference between the threshold and the peak of the AP. Input resistance was calculated using the change of voltage in response to the negative -20 pA current injection. Rheobase was approximated by taking the largest current injection step that did not evoke APs. The relationship between current injection and AP frequency was assessed by 1-second current injections of magnitudes ranging from -100 pA to 500 pA with a 1.5 second intersweep interval. APs were counted only if they reached a peak value of at least -10 mV. Neurons which had resting membrane potentials more depolarized than -65 mV (but less than -50 mV) were injected with DC current to bring to -65 mV to compare excitability across all cells.

To examine synaptic neurotransmission between PV-INs and excitatory cells, pairs of nearby cells (<100 μ m inter-soma distance) were simultaneously recorded in the whole-cell configuration similar to our previous studies (Kaneko et al., 2022 [DOI](#); Feng et al., 2024 [DOI](#)). Monosynaptic connections were determined by stimulating APs in one neuron via depolarizing current injections and looking for unitary post-synaptic potentials in the other cell. Potential connections were confirmed by switching the postsynaptic cell to voltage-clamp (holding potential, -70 mV). Recordings of unitary synaptic currents were obtained by stimulating the presynaptic cell with trains of 1-ms square wave pulses of 2 nA at frequencies of 5, 10, 20, 40, 80, and 120 Hz (for PV-INs only). Unitary synaptic currents were detected if their amplitude exceeded 10 pA and were otherwise considered a failure event. Connection probabilities were compared using Fisher's Exact test. Synaptic latency was calculated by taking the time difference between the peak of the AP and the onset of the evoked uIPSC.

All electrophysiology analysis was completed using custom-written MATLAB analysis scripts and/or confirmed manually using ClampFit (Molecular Devices). Statistical comparisons for physiological measures were completed using either unpaired t-tests or repeated-measures Two-way ANOVA followed by Sidak's multiple comparisons test. Significance was determined and denoted as * $P < 0.05$, ** $P < 0.01$, and *** $P < 0.001$.

Data availability

Electrophysiological data has been deposited at: <https://gin.g-node.org/GoldbergNeuroLab> [DOI](#). The code for the analyses presented in this paper is openly accessible at: <https://github.com/orgs/GoldbergNeuroLab> [DOI](#).

Figure supplements

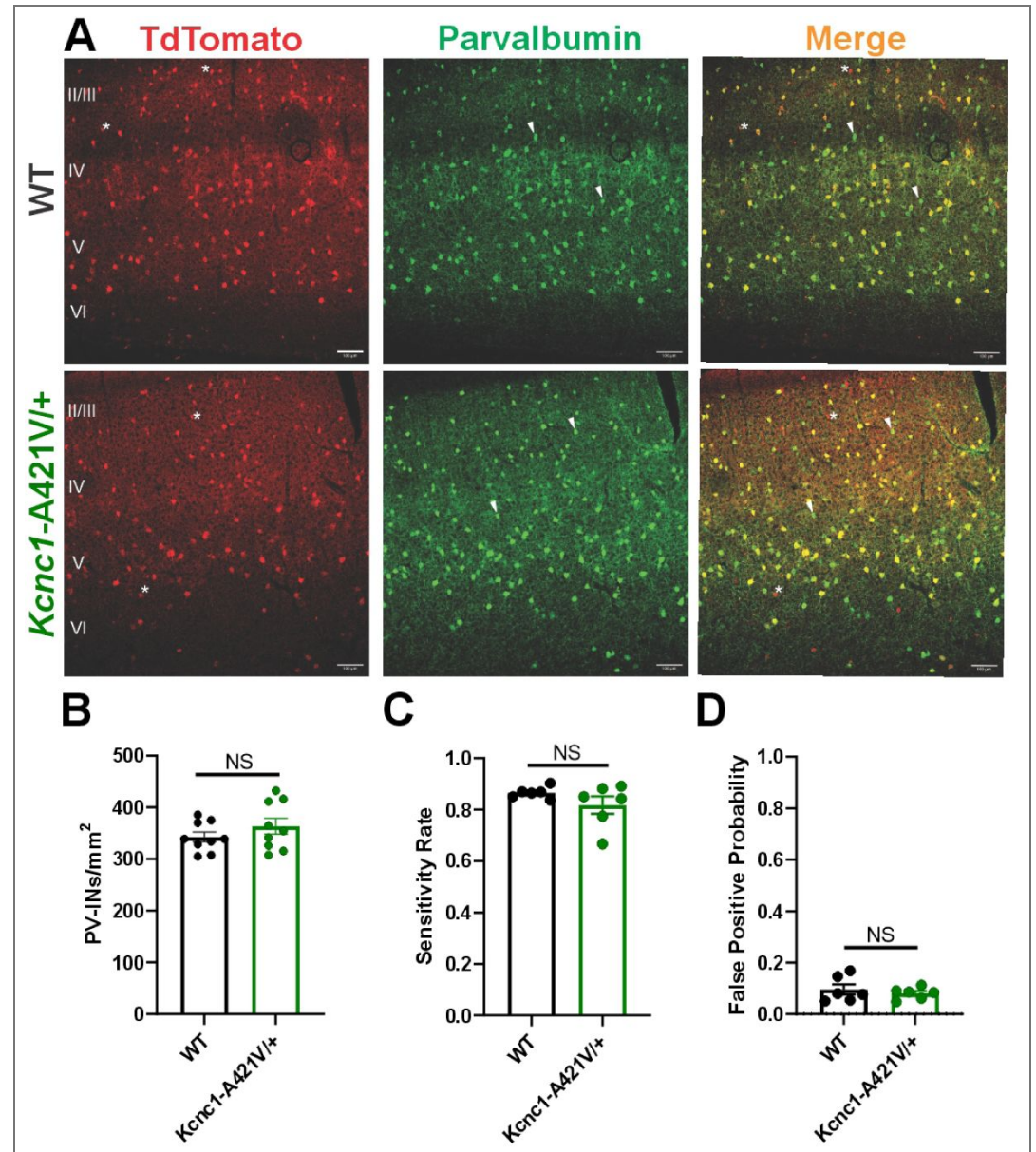


Figure 1—figure supplement 1. Pvalb-TdTomato reporter effectively labels PV-INs in WT and *Kcnc1-A421V/+* mice. **A.** Representative immunohistochemistry images for parvalbumin in WT (top row) and *Kcnc1-A421V/+* (bottom row) mice (P21-33) showing a high degree of overlap between the tdTomato reporter and parvalbumin expression. The asterisk indicates rare cells that are tdTomato+, but parvalbumin-. The arrowhead indicates cells that are parvalbumin+ but tdTomato-. Scale bar, 100μm. **B.** Counts of PV cells per unit area (mm²) are not different between WT and *Kcnc1-A421V/+* mice (N=3 mice/genotype). **C.** Sensitivity rate (probability of a cell being tdTomato+ if it is parvalbumin+; N=6 mice/genotype). **D.** False-positive probability (proportion of all tdTomato+ cells that are parvalbumin negative; N=6 mice/genotype).

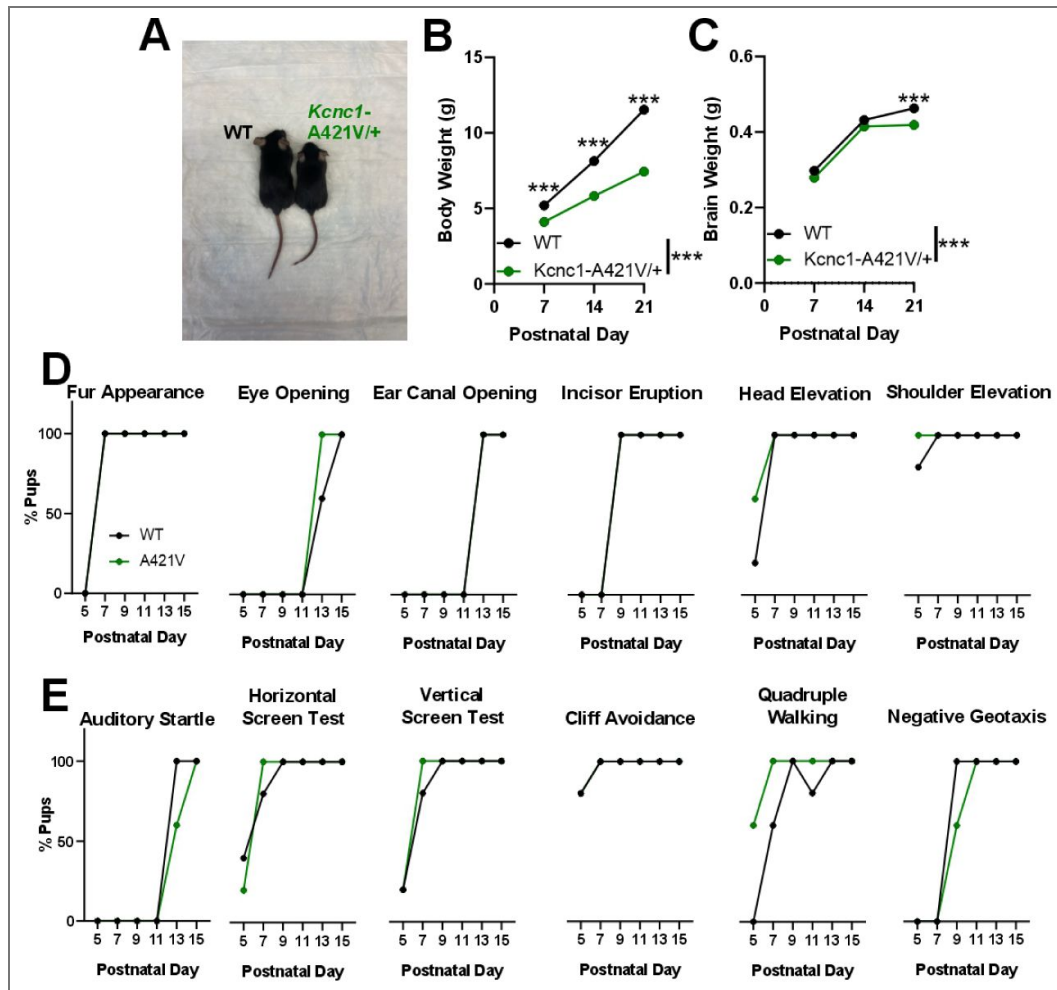


Figure 1—figure supplement 2. Early postnatal development of *Kcnc1*-A421V/+ mice.

A. Representative example image showing littermate WT and *Kcnc1*-A421V/+ mice at postnatal day 21. **B.** Average body weights for WT (N=10; black) and *Kcnc1*-A421V/+ (N=11; green) at postnatal days 7, 14, and 21. **C.** Average brain weights for WT and *Kcnc1*-A421V/+ mice at postnatal day 7 (WT, N = 2; *Kcnc1*-A421V/+, N = 6), 14 (WT, N=3; *Kcnc1*-A421V/+, N=3), and 21 (WT, N=4; *Kcnc1*-A421V/+, N=6). Note that for both B and C, error bars depicting SEM are present, but do not appear beyond the symbols. **D.** Onset of developmental and motor benchmarks for WT (N=7) and *Kcnc1*-A421V/+ (N=10) mice. **E.** Onset of startle reflex and other typical behavioral/motor milestones in WT (N=7) and *Kcnc1*-A421V/+ (N=10) mice. ***P<0.001 by Mixed-Effects Analysis followed by Sidak's Multiple comparisons post-hoc test.

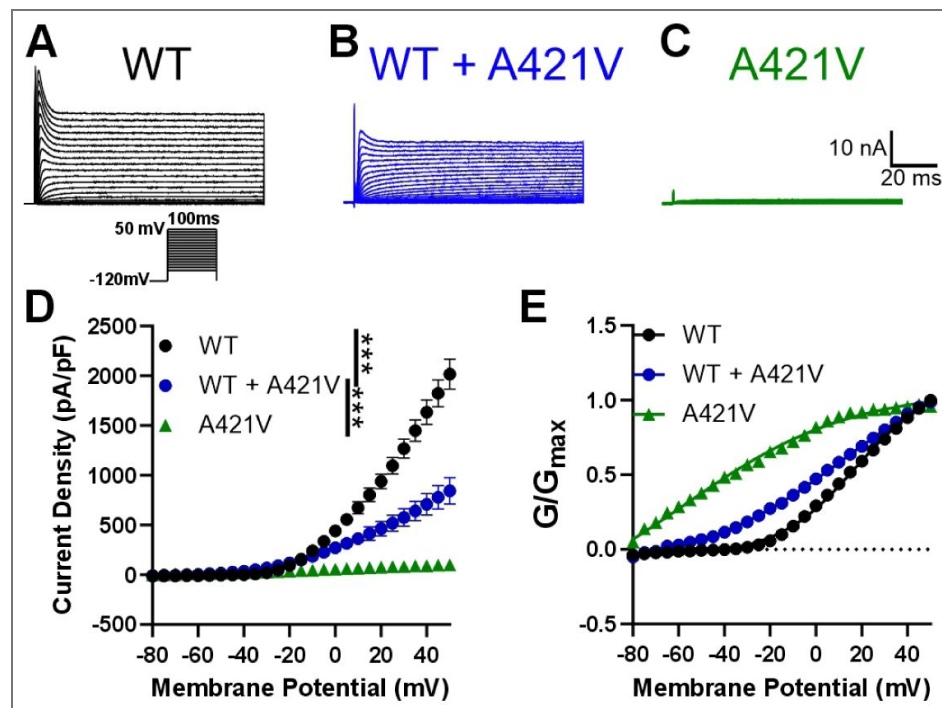


Figure 3—figure supplement 1. Loss of potassium current density in A421V-expressing HEK cells.

A-C. Example Kv3.1 currents in HEK cells expressing WT (A; black), a 50:50 mixture of WT and A421V (B; blue), and A421V (C; green) Kv3.1 subunits. Voltage command protocol shown below WT example. **D.** Average current density for WT (black), WT + A421V (blue), and A421V (green). Note that the A421V variant leads to a profound loss of function. **E.** Average normalized conductance relative to membrane potential for WT (black), WT + A421V (blue), and A421V (green). G-V curves for the A421V variant should be interpreted with caution as the recorded currents are exceedingly small and cannot be easily differentiated from the small endogenous delayed rectifier potassium currents known to be present in HEK cells.

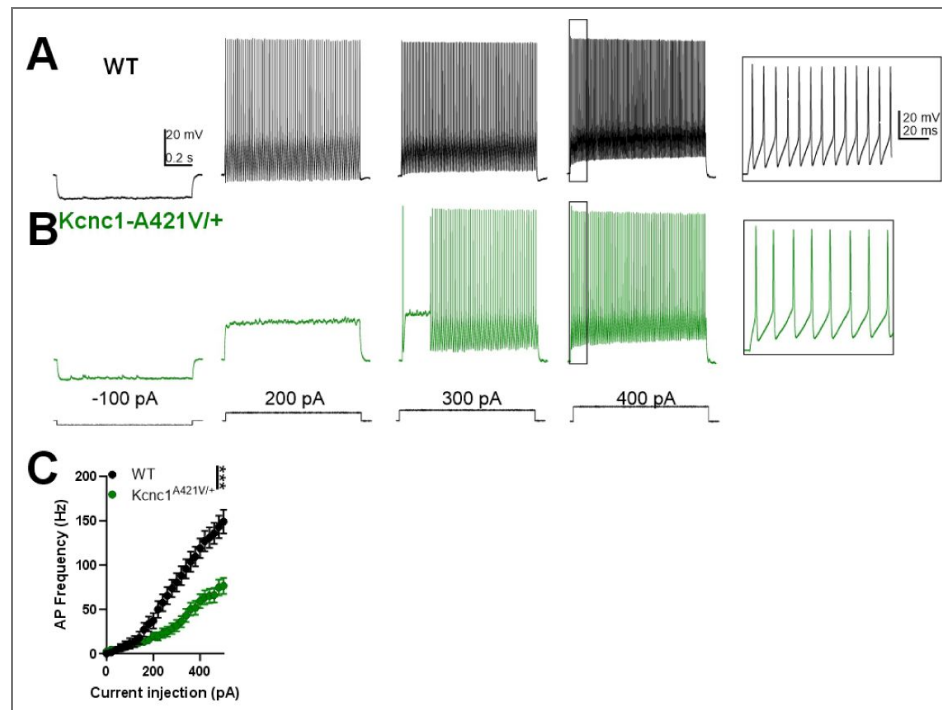


Figure 4—figure supplement 1. Intrinsic physiology of neocortical layer II-IV PV-INs remains impaired when resting membrane potential is not normalized.

(A-B) Representative example traces for juvenile (P16-21) WT (A, black) and *Kcnc1-A421V/+* (B, green) PV-INs generating APs at current injections of -100, 200, 300, and 400 pA from their resting membrane potential without DC bias current (as in Figure 3). C. Average frequency-current relationship for WT (n=20 cells, N=9 mice) and *Kcnc1-A421V/+* (n=32 cells, N=12 mice) PV-INs with uncorrected resting membrane potential. Data are shown as mean $\pm$ SEM and significance (***) $P < 0.001$ was determined using repeated measures Two-way ANOVA.

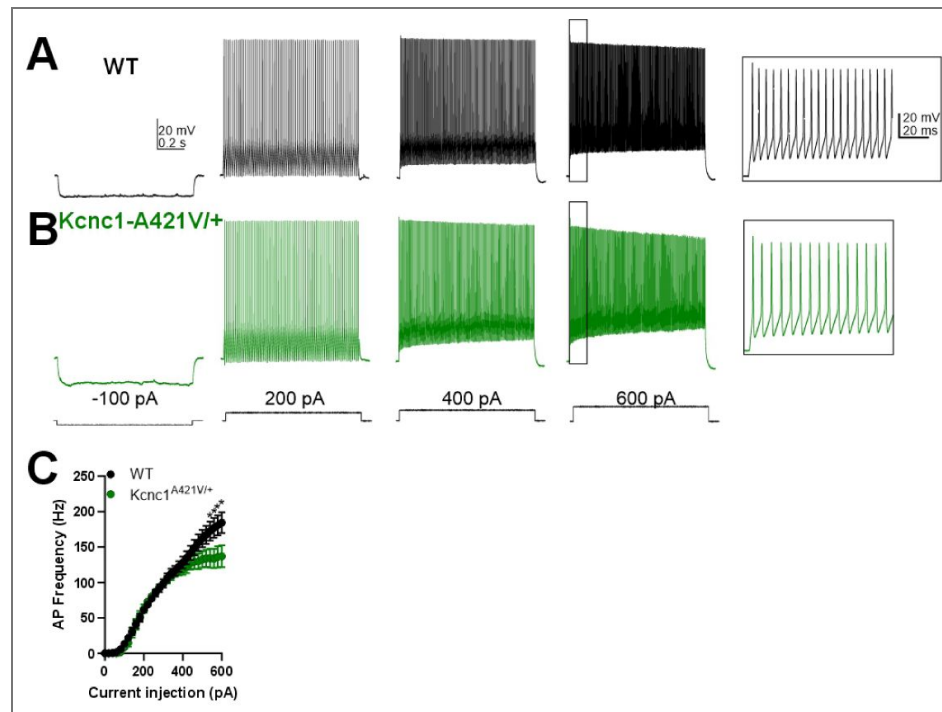


Figure 4—figure supplement 2. Subtle abnormalities in neocortical layer V PV-INs from juvenile (P16-21) *Kcnc1-A421V/+*.

(A-B) Representative example traces showing spiking in layer V neocortical PV-INs from WT (A, black) and *Kcnc1-A421V/+* (B, green) mice in response to depolarizing current injections. (C) Average PV-IN spiking frequency relative to current injection. A significant interaction effect between genotype and current injection was observed ($***P < 0.001$) by two-way ANOVA. Data are shown as mean $\pm$ SEM and significance of post-hoc comparisons ($*P < 0.05$) by repeated-measures Two-way ANOVA followed by Sidak multiple comparisons test.

Figure 4—figure supplement 3.

A. Representative images taken at 10X (left) and 40X (right) magnification of a PV-positive cell in the reticular thalamus recorded in the whole-cell configuration to characterize intrinsic excitability. **B.** Representative example traces of APs generated in response to current injections of varying magnitudes. **C.** Average hyperpolarization-induced rebound APs in WT ($n=16$ cells, $N=4$ mice) and *Kcnc1-A421V/+* ($n=19$ cells, $N=5$ mice) in response to various current injections from 0 to -100 pA. **D.** Frequency-current relationship shows impaired intrinsic excitability in the RT PV cells relative to depolarizing current injections (0-400 pA). Data are shown as mean $\pm$ SEM and significance is denoted as * $P<0.05$ or *** $P<0.001$ by Repeated-measures Two-way ANOVA.

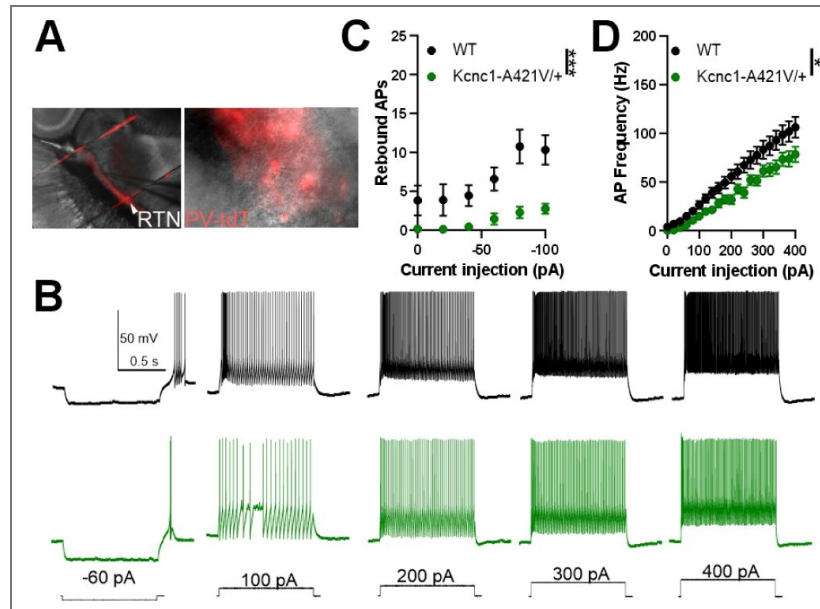
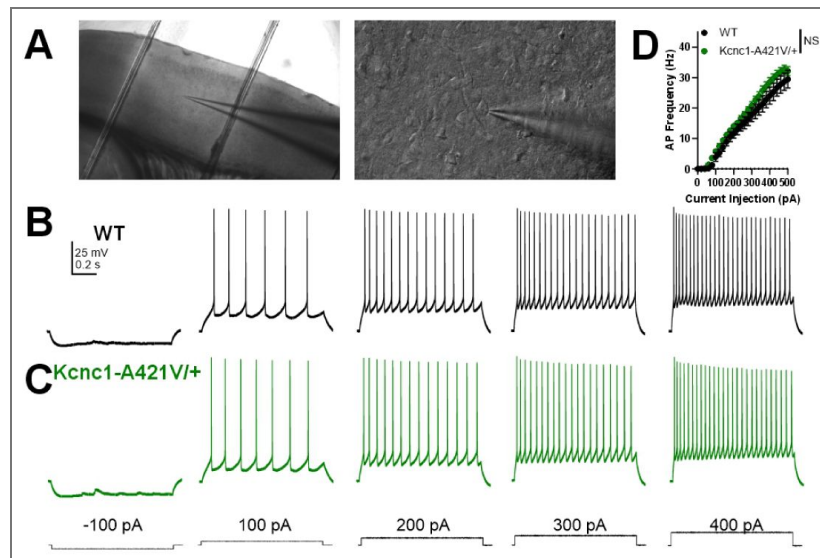


Figure 5—figure supplement 1. Intrinsic excitability is unchanged in excitatory cells from P32-42 *Kcnc1-A421V/+* mice.

(A) Representative images taken at 10X (left) and 40X (right) magnification of a recorded layer IV excitatory cell. **(B-C)** Representative example traces of evoked APs in response to depolarizing current injections in cortical layer IV excitatory neurons from WT (B) and *Kcnc1-A421V/+* (C) mice. **(D)** Average frequency-current relationship for WT (black; $n=12$, $N=4$) and *Kcnc1-A421V/+* ($n=12$, $N=4$) mice. Data are shown as mean $\pm$ SEM and all results failed to reach significance determined via repeated-measures Two-way ANOVA.



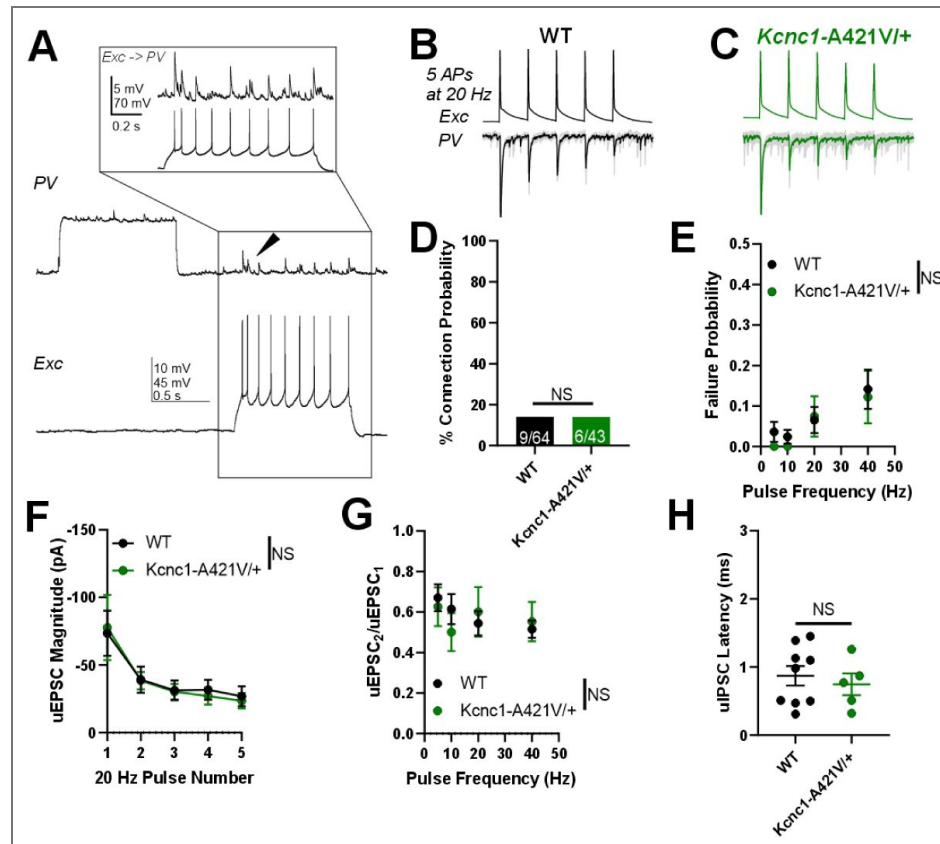


Figure 6—figure supplement 1. Excitatory neuron to PV-IN unitary excitatory synaptic neurotransmission is unaltered in juvenile (P16-21) *Kcnc1-A421V/+* mice.

Excitatory neuron to PV-IN excitatory synaptic neurotransmission is unaltered in *Kcnc1-A421V/+* mice. **A.** Example traces of simultaneous recordings of a synaptically-connected PV-IN and excitatory cell neuron pair in which generation of APs in the excitatory cell leads to uEPSPs in the PV-IN. The inset shows an expanded view of the uEPSPs in the PV-IN generated from each AP in the excitatory cell. **B-C.** Representative example traces of uEPSCs generated in WT (**B**) and *Kcnc1-A421V/+* pairs of excitatory cells (top trace) and nearby PV-INs (bottom trace). In response to a 20-Hz train of five APs generated in the excitatory cells, uIPSCs are recorded in the PV-INs (shown in gray) with the average of numerous sweeps shown for WT (black) or for *Kcnc1-A421V/+* (green). **D.** Connection probability between the excitatory cell and the PV-IN for WT ($n=9$ of 64 pairs from $N=7$ mice) and *Kcnc1-A421V/+* ($n=6$ of 43 pairs from $N=5$ mice). **E.** Average failure rate for WT and *Kcnc1-A421V/+* neuron pairs relative to presynaptic stimulation frequency. **F.** Average uEPSC magnitude in response to a train of five APs generated at 20 Hz in WT and *Kcnc1-A421V/+* neuron pairs. **G.** Paired-pulse ratio of the second uEPSC to the first uEPSC in WT and *Kcnc1-A421V/+* mice. **H.** Average synaptic latency between peak of AP to peak of uEPSC in both WT and *Kcnc1-A421V/+* mice.

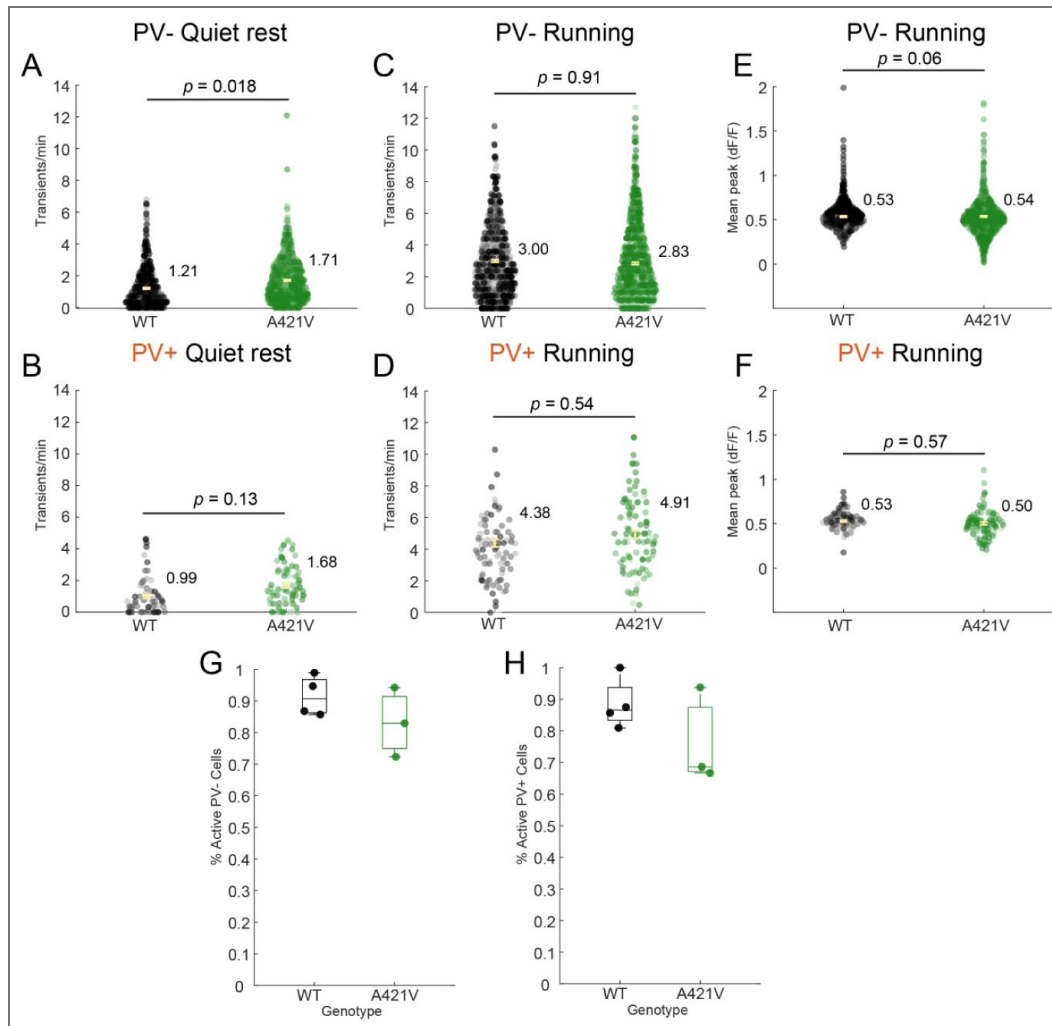


Figure 8—figure supplement 1.

A-B. Transients per minute during quiet rest in only active (A) PV-(WT, n=847 cells, N=4 mice; *Kcnc1*-A421V, n=901 cells, N=3 mice) and (B) PV+ cells (WT, n=65 cells; *Kcnc1*-A421V, n=60 cells). **C-D.** Transients per minute during running in (C) PV-(WT, n=885 cells, N=4 mice; *Kcnc1*-A421V, n=1041 cells, N=3 mice,) and (D) PV+ cells (WT, n=110 cells, N=4 mice; *Kcnc1*-A421V, n=100 cells, N=3 mice). **E-F.** Mean transient height in (E) PV- and (F) PV+ cells. Error bars indicate standard error of the mean, and values next to data points are the mean by genotype. Data points are shaded by mouse identity. Statistical comparisons were performed using mixed-effects modeling. **G-H.** Boxplot of percent of active (G) PV- and (H) PV+ cells for each mouse.

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Additional files

[Figure 8 video supplement 1](#) 

[Figure 9 video supplement 1](#) 

[Figure 9 video supplement 2](#) 

Additional information

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National Institute of Neurological Disorders and Stroke	F32NS126234	Eric R Wengert
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Peer reviews

Reviewer #1 (Public review):

Summary:

The authors have created a new model of KCNC1-related DEE in which a pathogenic patient variant (A421V) is knocked into mouse in order to better understand the mechanisms through which KCNC1 variants lead to DEE.

Strengths:

- (1) The creation of a new DEE model of KCNC1 dysfunction.
 - (2) InVivo phenotyping demonstrates key features of the model such as early lethality and several types of electrographic seizures.
 - (3) The ex vivo cellular electrophysiology is very strong and comprehensive including isolated patches to accurately measure K⁺ currents, paired recording to measure evoked synaptic transmission, and the measurement of membrane excitability at different timepoint and in two cell types.
 - (4) 2P imaging relates the cellular dysfunction in PV neurons to epilepsy.
- <https://doi.org/10.7554/eLife.103784.2.sa3>

Reviewer #2 (Public review):

Summary:

Wengert et al. generated and comprehensively characterized the *Kcnc1* A421V/+ knock-in mouse, which models developmental epileptic encephalopathy. The *Kcnc1* gene encodes the Kv3.1 channel subunit, which, similar to the role of BK-channels in some excitatory neurons, facilitates high-frequency firing in inhibitory neurons by accelerating the downward hyperpolarization of individual action potentials. Although various *Kcnc1* mutations are linked to developmental epileptic encephalopathies, the functional impact of the A421V mutation remained controversial. To elucidate its effect on the neuronal excitability and neurological functions, the authors generated cre-dependent KI mice and thoroughly characterized them using neonatal neurological assessments, high-quality in vitro electrophysiology, and in vivo imaging/electrophysiology analyses. These studies revealed impaired excitability in the PV⁺ inhibitory interneurons, correlating with the emergence of epilepsy and premature death. Overall, this study provides strong support for the role of the A421V mutation in disrupting inhibitory function.

Overall, the study is well-designed and conducted at a high quality. The use of a Cre-dependent KI system is effective for maintaining the mutant line despite the premature death phenotype, and may also minimize the phenotype drift that can arise when breeding from mice using milder phenotype manifestation (as ones with severe phenotype often fail to

reproduce). The neonatal behavior analysis is thoroughly conducted, and the in vitro electrophysiology studies are of high quality, providing robust insights into the functional impact of the mutation.

One limitation of this study is the demonstration of the trafficking defect of mutant Kv3.1, which relies solely on the fluorescence density, and such analysis often lacks a rigorous quantitative measurement. A biochemical analysis (surface biotinylation or immunoblot using membrane fractionation) will make the conclusion more convincing, although this poses a technical challenge as the Kv3.1 is expressed primarily expressed only in a subset of PV+ cells.

While the study focused on the superficial layer because Kv3.1 is the major channel subunit, some of the neurons co-express Kv3.2, and Kv3.1 and Kv3.2 can form heteromeric channels. It would be interesting to explore whether the mutant Kv3.1 subunits exert a dominant-negative effect on Kv3.2 in these populations.

<https://doi.org/10.7554/eLife.103784.2.sa2>

Reviewer #3 (Public review):

Summary:

Here Wengert et al., establish a rodent model of KCNC1 (Kv3.1) epilepsy by introducing the A421V mutation. The authors perform video-EEG, slice electrophysiology, and in vivo 2P imaging of calcium activity to establish a disease mechanisms involving impairment in the excitability of fast spiking parvalbumin (PV) interneurons in the cortex and thalamic PV cells.

Outside out nucleated patch recordings were used to evaluate the biophysical consequence of the A421V mutation on potassium currents and showed a clear reduction in potassium currents. Similarly action potential generation in cortical PV interneurons was severely reduced. Given that both potassium currents and action potential generation was found to be unaffected in excitatory pyramidal cells in the cortex the authors propose that loss of inhibition leads to hyperexcitability and seizure susceptibility in a mechanism similar to that of Dravet Syndrome.

Strengths:

This manuscript establishes a new rodent model of KCNC1-developmental and epileptic encephalopathy. The manuscript provides strong evidence that parvabumin interneurons are impaired by the Kcnc1-A421V mutation and that cortical excitatory neurons are not impaired. Together, these findings support the conclusion that seizure phenotypes associated with Kcnc1-A421V are caused by impaired cortical inhibition.

Weaknesses:

The manuscript identifies a partial mechanism of disease that leaves several aspects unresolved including the possible role of subcortical regions in the seizure mechanism. Similarly, while the authors identify a reduction in potassium currents and a reduction in PV cell surface expression of Kv3.1 why the A421V missense mutation leads to a more severe phenotype than previously reported loss-of-function mutations in Kv3.1is not clear.

<https://doi.org/10.7554/eLife.103784.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Summary:

The authors have created a new model of KCNC1-related DEE in which a pathogenic patient variant (A421V) is knocked into a mouse in order to better understand the mechanisms through which KCNC1 variants lead to DEE.

Strengths:

(1) The creation of a new DEE model of KCNC1 dysfunction.

(2) In Vivo phenotyping demonstrates key features of the model such as early lethality and several types of electrographic seizures.

(3) The ex vivo cellular electrophysiology is very strong and comprehensive including isolated patches to accurately measure K⁺ currents, paired recording to measure evoked synaptic transmission, and the measurement of membrane excitability at different time points and in two cell types.

We thank Reviewer 1 for these positive comments related to strengths of the study.

Weaknesses:

(1) The assertion that membrane trafficking is impaired by this variant could be bolstered by additional data.

We agree with this comment. However, given the technical challenges of standard biochemical experiments for investigating voltage-gated potassium channels (e.g., antibody quality), the lack of a Kv3.1-A421V specific antibody, and the fact that Kv3.1 is expressed in only a small subset of cells, we did not undertake this approach. However, we did perform additional experiments and analysis to improve the rigor of the experiments supporting our conclusion that membrane trafficking is impaired in the *Kcnc1-A421V/+* mouse.

Such experiments support a highly significant and robust difference in our (albeit imperfect) measurement of the membrane:cytosol ratio of Kv3.1 immunofluorescence between WT and *Kcnc1-A421V/+* mice, which is consistent with lack of membrane trafficking (Figure 3). In the revised manuscript, we have added additional data points to this plot and updated the representative example images using improved imaging techniques to better showcase how *Kcnc1-A421V/+* PV-INs differ from age-matched WT littermate controls. We think the result is quite clear. Future biochemical experiments perhaps best performed in a culture system *in vitro* could provide additional support for this conclusion.

(2) In some experiments details such as the age of the mice or cortical layer are emphasized, but in others, these details are omitted.

We apologize for this omission. We have now clarified the age of the mice and cortical layer for each experiment in the Methods and Results sections as well as figure legends.

*(3) The impairments in PV neuron AP firing are quite large. This could be expected to lead to changes in PV neuron activity outside of the hypersynchronous discharges that could be detected in the 2-photon imaging experiments, however, a lack of an effect on PV neuron activity is only loosely alluded to in the text. A more formal analysis is lacking. An important question in trying to understand mechanisms underlying channelopathies like KCNC1 is how changes in membrane excitability recorded at the whole cell level manifest during ongoing activity *in vivo*. Thus, the significance of this work would be greatly improved if it could address this question.*

Yes, the impairments in the neocortical PV-IN excitability are notably severe relative to other PV interneuronopathies that we and others have directly investigated (e.g., Kv3.1 or Kv3.2/-

knockout mice; *Scn1a*^{+/-} mice). In the revised version of the manuscript, we have now added a more thorough in vivo 2P calcium imaging investigation and analysis of our in vivo 2P calcium imaging data of PV-IN (and presumptive excitatory cell) neural activity (Figure 8 and Supplementary Figure 9, Methods- lines 230-271 Results- lines 630-657, and Discussion lines- 795-814).

Because of the prominent recruitment of neuropil during presumptive myoclonic seizures, further investigation of individual neuronal excitability in vivo required a slightly different labeling strategy now using a soma-tagged GCaMP8m as well as a separate AAV containing tdTomato driven by the PV-IN-specific S5E2 enhancer. Our new results reveal an increase in the baseline calcium transient frequency in non-PV-INS, and reduced mean transient amplitudes in both non-PV cells and PV-INS. These interesting findings, which are consistent with attenuated PV-IN-mediated perisomatic inhibition leading to disinhibited excitatory cells in the *Kcnc1*-A421V/+ mice, link our in vivo results to the slice electrophysiology experiments. Of course, there are residual issues with the application of this technique to interneurons and the ability to resolve individual or small numbers of spikes, which likely explains the lack of genotype difference in calcium transient frequency in PV-INS.

(4) Myoclonic jerks and other types of more subtle epileptiform activity have been observed in control mice, but there is no mention of littermate control analyzed by EEG.

We performed additional experiments as requested and did not observe myoclonic jerks or any other epileptic activity in WT control mice. We have included this data in the revised manuscript (Figure 9C).

Reviewer #2 (Public review):

Summary:

*Wengert et al. generated and thoroughly characterized the developmental epileptic encephalopathy phenotype of *Kcnc1*A421V/+ knock-in mice. The *Kcnc1* gene encodes the Kv3.1 channel subunit. Analogous to the role of BK channels in excitatory neurons, Kv3 channels are important for the recurrent high-frequency discharge in interneurons by accelerating the downward hyperpolarization of the individual action potential. Various *Kcnc1* mutations are associated with developmental epileptic encephalopathy, but the effect of a recurrent A421V mutation was somewhat controversial and its influence on neuronal excitability has not been fully established. In order to determine the neurological deficits and underlying disease mechanisms, the authors generated cre-dependent KI mice and characterized them using neonatal neurological examination, high-quality in vitro electrophysiology, and in vivo imaging/electrophysiology analyses. These analyses revealed excitability defects in the PV+ inhibitory neurons associated with the emergence of epilepsy and premature death. Overall, the experimental data convincingly support the conclusion.*

Strengths:

The study is well-designed and conducted at high quality. The use of the Cre-dependent KI mouse is effective for maintaining the mutant mouse line with premature death phenotype, and may also minimize the drift of phenotypes which can occur due to the use of mutant mice with minor phenotype for breeding. The neonatal behavior analysis is thoroughly conducted, and the in vitro electrophysiology studies are of high quality.

We appreciate these positive comments from Reviewer 2.

Weaknesses:

While not critically influencing the conclusion of the study, there are several concerns.

In some experiments, the age of the animal in each experiment is not clearly stated. For example, the experiments in Figure 2 demonstrate impaired K⁺ conductance and membrane localization, but it is not clear whether they correlated with the excitability and synaptic defects shown in subsequent figures. Similarly, it is unclear how old mice the authors conducted EEG recordings, and whether non-epileptic mice are younger than those with seizures.

We have now updated the manuscript to include clear report of age for all experiments including the impaired K⁺ conductance (now Figure 3) and EEG (now Figure 9). There was no intention to omit this information. The recordings of K⁺ conductance impairments in PV-INs from Kcnc1-A421V/+ mice were completed at P1621. Thus, we interpret the loss of potassium current density to be causally linked with the impairments in intrinsic physiological function at that same time-period in neocortical layer II-IV PV-INs and more subtly in PV-positive cells in the RTN and neocortical layer V PVINs.

Mice used in the EEG experiments were P24-48, an age range which roughly corresponded with the midpoint on the survival curve for Kcnc1-A421V/+ mice. Although we saw significant mouse-to-mouse variability in seizure phenotype, no Kcnc1-A421V/+ mice completely lacked epilepsy or marked epileptiform abnormalities, neither of which were seen in WT mice. We did not detect a clear relationship between seizure frequency/type and mouse age.

The trafficking defect of mutant Kv3.1 proposed in this study is based only on the fluorescence density analysis which showed a minor change in membrane/cytosol ratio. It is not very clear how the membrane component was determined (any control staining?). In addition to fluorescence imaging, an addition of biochemical analysis will make the conclusion more convincing (while it might be challenging if the Kv3.1 is expressed only in PV+ cells).

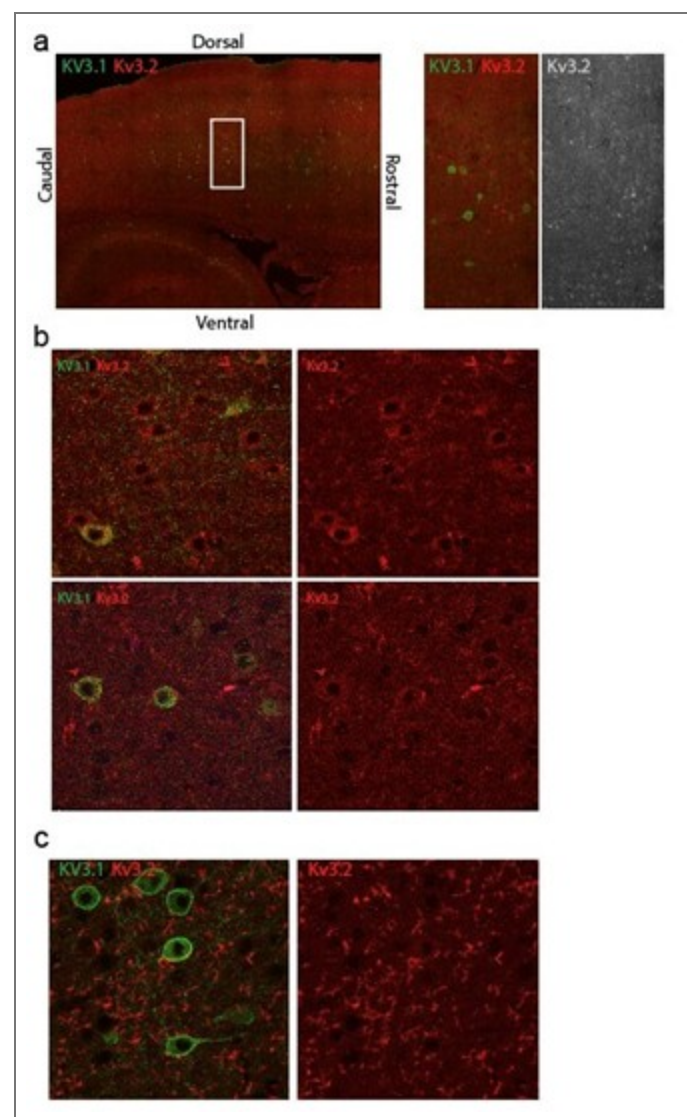
This relates to comment 3 of Reviewer 1. We agree that, in the initial submission of the manuscript, the evidence from IHC for Kv3.1 trafficking deficits was somewhat subtle. In the revised version of the paper, we have gathered additional replicates of this original experiment with improved imaging quality and clarify how the membrane component was specified, to now show a robust and highly significant (**P<0.001) decrease in membrane:cytosol Kv3.1 ratio. We have also now provided new example images better showcasing the deficits observed in the Kcnc1-A421V/+ mice (Figure 3). The membrane compartment was defined as the outermost 1 micron of the parvalbumin-defined cell soma (drawn blind to the Kv3.1b signal), and, importantly, all analysis was conducted blinded to mouse genotype. These measures help to ensure that the result is robust and unbiased. Nonetheless, we have added a paragraph in the Discussion section highlighting the limitations of our IHC evidence for trafficking impairment (Lines 868-883).

While the study focused on the superficial layer because Kv3.1 is the major channel subunit, the PV+ cells in the deeper cortical layer also express Kv3.1 (Chow et al., 1999) and they may also contribute to the hyperexcitable phenotype via negative effect on Kv3.2; the mutant Kv3.1 may also block membrane trafficking of Kv3.1/Kv3.2 heteromers in the deeper layer PV cells and reduce their excitability. Such an additional effect on Kv3.2, if present, may explain why the heterozygous A421V KI mouse shows a more severe phenotype than the Kv3.1 KO mouse (and why they are more similar to Kv3.2 KO). Analyzing the membrane excitability differences in the deep-layer PV cells may address this possibility.

We appreciate this thoughtful suggestion. We have now provided data from neocortical layer V PV interneurons in the revised manuscript (Supplementary Figure 5). Abnormalities in intrinsic excitability from neocortical layer V PV-INs in Kcnc1A421V/+ mice were present, but less pronounced than in PV-INs from more superficial cortical layers. These results are

consistent with the view that greater relative expression of Kv3.2 “dilutes” the impact of the Kv3.1 A421V/+ variant. More specific determination of whether the A421V/+ variant impairs membrane trafficking and/or gating of Kv3.2 remains unclear.

We attempted to assess how the mutant Kv3.1 affects Kv3.2 localization, but were unsuccessful due to the lack of reliable antibodies. After immunostaining mouse brain sections with two different anti-Kv3.2 antibodies, only one produced somewhat promising signal (see below). However, even in this case, Kv3.2 staining was successful only once (out of five independent staining experiments) and the signal varied across cortical regions, showing widespread cellular Kv3.2 signal in some areas (b, top panel), and barely detectable signal in others, regardless of Kv3.1 expression. In the remaining four attempts, we detected only ‘fiber-like’ immunostaining signal, further diminishing our confidence in anti-Kv3.2 antibody, although results could be improved with still further testing and refinement which we will attempt. Consequently, this important question remains unsolved in this study.



Author response image 1. Immunostaining of Kv3.1 and Kv3.2 in sagittal mouse brain sections. a) An example of intracellular Kv3.2 immunostaining signal, variable across the cortex of a WT mice independent of Kv3.1 expression b) Kv3.2 is detectable intracellularly in most of the cells in the top panel but barely detectable in the lowest panel. c) Representative image of Kv3.2 immunostaining signal in other sagittal mouse brain sections.

We have discussed these important implications and limitations of our results in the Discussion (Lines 868-883). We agree with the Reviewer's interpretation that an impact on Kv3.1/Kv3.2 heteromultimers across the neocortex may explain why the Kcnc1A421V/+ mouse exhibits a more severe phenotype than Kv3.1-/- or Kv3.2-/- mice (see below), a view which we have attempted to further clarify in the Conclusion.

In Table 1, the A421V PV+ cells show a depolarized resting membrane potential than WT by ~5 mV which seems a robust change and would influence the circuit excitability. The authors measured firing frequency after adjusting the membrane voltage to -65mV, but are the excitability differences less significant if the resting potential is not adjusted? It is also interesting that such a membrane potential difference is not detected in young adult mice (Table 2). This loss of potential compensation may be important for developmental changes in the circuit excitability. These issues can be more explicitly discussed.

We do not entirely understand this finding and its apparent developmental component. It could be compensatory, as suggested by the Reviewer; however, it is transient and seems to be an isolated finding (i.e., it is not accompanied by compensation in other properties). It is also possible that this change in Kcnc1-A421V/+ PV-INs may reflect impaired/delayed development. We cannot test excitability at a meaningfully later time point as the mice are deceased.

The revised version of the manuscript contains additional data (Supplementary Figure 4) showing that major deficits in intrinsic excitability are still observed even when the resting membrane potential is left unadjusted. These results are further discussed in the Results section (lines 522-523) and the Discussion section (lines 727-731).

Reviewer #3 (Public review):

Summary:

Here Wengert et al., establish a rodent model of KCNC1 (Kv3.1) epilepsy by introducing the A421V mutation. The authors perform video-EEG, slice electrophysiology, and in vivo 2P imaging of calcium activity to establish disease mechanisms involving impairment in the excitability of fast-spiking parvalbumin (PV) interneurons in the cortex and thalamic PV cells.

Outside-out nucleated patch recordings were used to evaluate the biophysical consequence of the A421V mutation on potassium currents and showed a clear reduction in potassium currents. Similarly, action potential generation in cortical PV interneurons was severely reduced. Given that both potassium currents and action potential generation were found to be unaffected in excitatory pyramidal cells in the cortex the authors propose that loss of inhibition leads to hyperexcitability and seizure susceptibility in a mechanism similar to that of Dravet Syndrome.

Strengths:

This manuscript establishes a new rodent model of KCNC1-developmental and epileptic encephalopathy. The manuscript provides strong evidence that parvalbumin-type interneurons are impaired by the A421V Kv3.1 mutation and that cortical excitatory neurons are not impaired. Together these findings support the conclusion that seizure phenotypes are caused by reduced cortical inhibition.

We thank Reviewer 3 for their view of the strengths of the study.

Weaknesses:

The manuscript identifies a partial mechanism of disease that leaves several aspects unresolved including the possible role of the observed impairments in thalamic neurons in the seizure mechanism. Similarly, while the authors identify a reduction in potassium currents and a reduction in PV cell surface expression of Kv3.1 it is not clear why these impairments would lead to a more severe disease phenotype than other loss-of-function mutations which have been characterized previously. Lastly, additional analysis of videoEEG data would be helpful for interpreting the extent of the seizure burden and the nature of the seizure types caused by the mutation.

We agree with this comment(s) from Reviewer 3. We studied neurons in the reticular thalamus and layer V neocortical PV-INs since they are also linked to epilepsy pathogenesis and are known to express Kv3.1. However, for most of the study, we focused on neocortical layer II-IV PV-INs, because these cells exhibited the most robust impairments in intrinsic excitability. Cross of our novel *Kcnc1-Flox(A421V)/+* mice to a cerebral cortex interneuron-specific driver that would avoid recombination in the thalamus, such as *Ppp1r2-Cre* (*RRID:IMSR_JAX:012686* <https://doi.org/10.1093/rrid/rrz012>), could assist in determining the relative contribution of thalamic reticular nucleus dysfunction to overall phenotype as used by (Makinson et al., 2017) to address a similar question; however, we have been unable to obtain this mouse despite extensive effort. There are of course other Kv3.1-expressing neurons in the brain, including in the hippocampus, amygdala, and cerebellum, and we have provided additional discussion (Lines 731-736) of this issue.

We further agree with the Reviewer that a major question in the field of KCNC1-related neurological disorders is the mechanistic underpinning of why the KCNC1-A421V variant leads to a more severe disease phenotype than other loss of function KCNC1 variants, and, further, why the mouse phenotype is more severe than the *Kcnc1* knockout. Previous results and our own recordings in heterologous systems suggest that the A421V variant is more profoundly loss of function than the R320H variant (Oliver et al., 2017; Cameron et al., 2019; Park et al., 2019), which is consistent with A421V having a more severe disease phenotype. Relative to knockout of Kv3.1, our results are consistent with the view that the A421V exhibits dominant negative activity by reducing surface expression of Kv3.1 and/or Kv3.2 (an effect that would not occur in knockout mice), with a possible additional contribution of impairing gating of those Kv3.1-A421V variant containing Kv3.1/Kv3.2 heteromultimers by inclusion of A421V subunits into the heterotetramer. Our finding that the magnitude of total potassium current was reduced in PV-INs by ~50% is consistent with a combination of these various mechanisms but does not distinguish between them.

In the revised version of the manuscript, we have provided a more complete discussion of these important remaining questions regarding our interpretation of how the severity of KCNC1 disorders relates to the biophysical features of the ion channel variant (lines 868883).

Recommendations for the authors

Reviewer #1 (Recommendations for the authors):

Major

*(1) The authors suggest that the reduced K⁺ current density in *Kcnc1-A421V/+* neurons is due in part to impaired trafficking and cell surface expression of Kv3.1 in these neurons. The data supporting this claim aren't completely convincing. First, it's difficult to visualize a difference in Kv3.1 localization in the images shown in panel H, and importantly, it seems problematic that the method to assess Kv3.1 levels in membrane vs. cytosol relied on using PV co-staining to define the membrane compartment as the outermost 1 μm of the PV-defined cell soma. This doesn't seem to be the best method to define the membrane compartment, as the PV signal should be largely cytosolic.*

As noted above, we have completed additional data collection to confirm our results, and have performed additional imaging and updated our example images to be more representative of the observed deficits in membrane Kv3.1 expression in the *Kcnc1-A421V/+* mice. We attempted to identify a marker to more clearly label the membrane to combine with PV immunocytochemistry but were unable to do so despite some effort.

*Is it possible that in control neurons, the cytosolic PV signal localizes within the membrane-bound Kv3.1 signal, with less colocalization, whereas in *Kcnc1-A421V/+* neurons, there would be more colocalization of the cytosolic PV and improperly trafficked Kv3.1.? Could the data be presented in this way showing altered colocalization of Kv3.1 with PV?*

We do not entirely understand the nature of this concern. In our experiments, we utilized the PV signal to determine the cell membrane and cytosolic compartments in an unbiased manner using a 1-micron shell traced around/outside the edge of the PV signal to define the membrane compartment, with the remainder of the area (minus the nuclear signal defined by DAPI) defined as the cytosol (see Methods 176-186). Because we did not identify any alterations in PV signal or correlation between PV immunohistochemistry and tdTomato expression in Cre reporter strains between WT and *Kcnc1-A421V/+* mice, we believe that our strategy for determining membrane:cytosol ratio of Kv3.1 in an unbiased manner is acceptable (albeit of course imperfect).

*Alternatively, membrane fractionation could be performed on WT vs *Kcnc1-A421V/+* neurons, followed by Western blotting with a Kv3.1 antibody to show altered proportions in the cytosolic vs. membrane protein fractions. It's important that these results are convincing, as the findings are mentioned in the Abstract, the Results section, and multiple times in the Discussion, although it is still unclear how much the potential altered trafficking contributes to the decrease in K⁺ currents versus changes in channel gating.*

Multiple technical barriers made it difficult for us to gain direct biochemical evidence for altered trafficking of the A421V/+ Kv3.1 variant (see above). It is not clear how membrane fractionation techniques could be easily applied in this case (at least by us) when PV-INs constitute 3-5% of all neocortical neurons. We further agree (as noted above) that it is difficult to properly disentangle the relative roles of impaired membrane trafficking vs. gating deficits to the observed effect; however, we think that both phenomena are likely occurring. In the revised version of the manuscript, we have more explicitly discussed these limitations in the Discussion section (Lines 868-883).

(2) More information is needed regarding the age of mice used for experiments for the following results (added to the Results section as well as figure legends):

PV density (Supplementary Figure 1)

K⁺ current data (Figure 2A-G)

Kv3.1 localization (Figure 2H and I)

RTN electrophysiology (Supplementary Figure 3)

Excitatory neuron electrophysiology (Figure 4)

In vivo 2P calcium imaging (Figure 7)

Video-EEG (Figure 8)

We apologize for omitting this critical information. In the revised manuscript, we have provided the age of mice for each of our experiments in the results section, in the figure legend, and in the methods section.

(3) It's unclear why developmental milestones/behavioral assessments were only done at P5-P10. In the previous publication of another Kcnc1 LOF variant (Feng et al. 2024), no differences were found at P5-P10, and it was suggested in the discussion that this finding was "consistent with the known developmental expression pattern of Kv3.1 in mouse, where Kv3.1 protein does not appear until P10 or later". In that paper, they did find behavioral deficits at 2-4 months. Even though this model is more severe than the previous model, it would be interesting to determine if there are any behavioral deficits at a later time point (especially as they find more neurophysiological impairments at P32-P42).

As in our previous study, the lack of clear behavioral deficits in developmental milestones from P5-15 is potentially expected considering the developmental expression of Kv3.1, and we performed these experiments primarily to showcase that the Kcnc1-A421V/+ mice exhibit otherwise normal overall early development (although this could be an artifact of the sensitivity of our testing methods).

For the revised manuscript, we have conducted additional experiments to investigate behavioral deficits in adult Kcnc1-A421V/+ mice. We found cognitive/learning deficits in both Kcnc1-A421V/+ mice relative to WT in both the Barnes maze (Figure 2A-C) and Ymaze (Figure 2D-F). Other aspects of animal behavior including cerebellar-related motor function are likely also impaired at post-weaning timepoints, and will be included in a forthcoming research study focusing on the motor function in these mice.

(4) In the Results section, it should be more clearly stated which cortical layer/layers are being studied. In some cases, it mentions layers 2-4, and in some, only layer 4, and in others, it doesn't mention layers at all. Toward the beginning of the Results section, the rationale for focusing on layers 2-4 to assess the effects of this variant should be well described and then, for each experiment, it should be stated which cortical layers were assessed. Related to this point, it seems electrophysiology was only done in layer 4; the rationale for this should also be included.

We have now clarified which neocortical layers were under investigation in the study. All PV-INs were targeted in somatosensory layers II-IV, while excitatory neurons were either cortical layer IV spiny stellate cells or pyramidal cells. Paired recordings were also completed in layer IV. We have also more explicitly articulated our rationale for looking at PV-INs in layers II-IV to examine the cellular/circuit-level impact of Kv3.1 in a model of developmental and epileptic encephalopathy (Lines 487-491).

(5) Kcnc1-A421V/+ PV neurons showed more robust impairments in AP shape and firing at P32-42 than at P16-21 (Figure 3), and only showed synaptic neurotransmission alterations at P32-42 (Figure 6). Thus, it's unclear why Kcnc1-A421V/+ excitatory neurons were only assessed at P16-21 (Figure 4 and Supplementary Figure 4 related to Figure 5), particularly if only secondary or indirect effects on this population would be expected.

We appreciate this excellent point raised by the Reviewer and we have taken the suggestion to examine excitatory neurons at P32-42 in addition to the earlier juvenile timepoint. Our new results from the later timepoint are similar to our results at P16-21: Excitatory neurons show no statistically significant impairments in intrinsic excitability at either of the two timepoints examined (Supplementary Figure 7). This adds support to our original conclusion that PV-INs represent the major driver of disease pathology across development.

(6) The 2P calcium imaging experiments are potentially interesting, however, a relationship between these results and the electrophysiology results for PV neurons is lacking. Was there an attempt to assess the frequency and/or amplitude of calcium events specifically in PV neurons, outside of the hypersynchronous discharges, to determine whether there are differences between WT and Kcnc1-A421V/+, as was seen in the electrophysiological analyses? It does seem there are some key differences between the two experiments (age: later timepoint for 2P vs. P16-21 and P32-42, layer: 2/3 vs. 4, and PV marking method: virus vs. mouse line), but the electrophysiological differences reported were quite strong. Thus, it would be surprising if there were no alterations in calcium activity among the Kcnc1-A421V/+ PV neurons.

In our initial experiments, the prominent neuropil GCaMP signal in Kcnc1-A421V/+ mice rendered it difficult to distinguish and accurately describe baseline neuronal excitability in PV-INs and non-PV cells. In our revised manuscript, we utilized a soma-tagged GCaMP8m and separately labeled PV-INs through S5E2-tdTomato. This strategy made it possible to assess the amplitude and frequency of calcium transients in both PV-positive and PV-negative cells in vivo. We have updated the description of our methods (lines 230-271) and our results (lines 630-657) in the revised manuscript.

As noted above, our more detailed analysis of somatic calcium transients in PV-IN and non-PV cells during quiet rest (Figure 8 and Supplementary Figure 9) shows that PV-INs from Kcnc1-A421V/+ mice are abnormally excitable- having reduced transient amplitude relative to WT controls. Interestingly, non-PV cells also exhibited an increased calcium transient frequency and reduced amplitude which is potentially consistent with reduced perisomatic inhibition causing disinhibition in cortical microcircuits. We again highlight that the slow kinetics of GCaMP combined with the calcium buffering and brief spikes of PVINs render quantification of action potential frequency and comparisons between groups difficult.

(7) As mentioned above, it would be helpful to state the time points or age ranges of these experiments to better understand the results and relate them to each other. For example, the 2P imaging showed apparent myoclonic seizures in 7/7 Kcnc1-A421V/+ mice (recorded for a total of 30-50 minutes/mouse), but the video-EEG showed myoclonic seizures in only 3/11 Kcnc1-A421V/+ mice (recorded for 48-72 hours/mouse). Were these experiments done at very different age ranges, so this difference could be due to some sort of progression of seizure types and events as the mice age? Is it possible these are not the same seizure types (even though they are similarly described)? This discrepancy should be discussed.

Mice in the EEG experiments were between the ages of P24 and 48, slightly younger than the age in which we carried out the in vivo calcium imaging experiments (>P50). Therefore, an age-related exacerbation in myoclonic jerks is possible.

As is highlighted by the Reviewer, it is interesting that the myoclonic seizures were only detected in a portion of the Kcnc1-A421V/+ mice during EEG monitoring (4/12). We believe that the difference is most likely driven by more sensitive detection of the myoclonic jerk activity and behavior in the 2P imaging of neuropil cellular activity compared to our video-EEG monitoring and 2P imaging of soma-tagged GCaMP. We have occasionally observed repetitive myoclonic jerking in mice that appears highly localized (i.e. one forepaw only) suggesting that the myoclonic seizures exist on a spectra of severity from focal to diffuse. It is therefore possible that myoclonic events and electrographic activity may be slightly underestimated in our video-EEG experiments?

We have now added a few lines discussing this discrepancy in the Discussion (lines 809814).

(8) Myoclonic jerks and other types of more subtle epileptiform activity have been observed in control mice. Was video-EEG performed on control mice? These data should

be added to Figure 8.

We have added recordings in control WT mice (N=4). We did not detect myoclonic jerks or other epileptiform activity in the control mice (Figure 9).

Minor

(1) In the first Results section, Line 365, the P value ($P < 0.001$) is different from that in the legend for Figure 1, line 743 ($P < 0.0001$).

We have fixed this discrepancy.

(2) For Supplementary Figure 1, it would be helpful to show images that span the cortical layers (1-6), as PV and Kv3.1 are both expressed across the cortical layers.

We have updated Supplementary Figure 1 with better example images that span the cortical layers.

(3) Error bars should be added to the line graphs in Supplementary Figure 2, particularly panels B and C. Some of the differences appear small considering the highly significant p-values (i.e. body weight at P7 and brain weight at P21).

The values shown in Supplementary Figure 2D-E are percentages of mice displaying a particular characteristic, so there is no variance for the data.

Supplementary Figure 2B-C actually do contain error bars plotted as SEM, however, because of the large number of N and small degree of variance in the measurements, the error bars are not apparent in the graphs. This has been noted in the Supplementary Figure 2 legend for clarity.

(4) In Figure 3, although the *Kcnc1-A421V/+* neurons have elevated AP amplitudes relative to WT, the representative traces for P16-21 and P32-42 groups appear strikingly opposite (traces in B in G appear to have much higher amplitudes than those in C and H). As this is one of the three AP phenotypes described, it would be nice to have it reflected in the traces.

We have updated our example traces to better represent our main findings including AP amplitude for both P16-21 and P32-42 timepoints.

(5) Were any effects on the AHP assessed in the electrophysiology experiments? As other studies have reported the effects of altered Kv3 channel activity on AHP, this parameter could be interesting to report as well.

We have now provided data on the afterhyperpolarization for each condition displayed in the Supplementary data tables. Interestingly, we failed to detect significant differences in AHP between WT and *Kcnc1-A421V/+* PV-INs, RTN neurons, or pyramidal cells, although we did identify differences in the dV/dt of the repolarization phase of the AP.

(6) The figure legend for Figure 7 has errors in the panel labeling (D instead of C, and two Fs).

This error has been corrected in the revised manuscript.

Reviewer #3 (Recommendations for the authors):

Specific comments and questions for the authors:

(1) Do the authors provide a reason for why the juvenile animals are unaffected by the A421V mutation? Is it that PV cells have not fully integrated at this early time point or

that Kv3.1 expression is low? Is the developmental expression profile of Kv3.1 in PV cells known and if so could the authors update the discussion with this information?

We interpret the normal early developmental milestones (P5-P15) to reflect that *Kcnc1-A421V/+* mice exhibit the onset of their neurological impairment at the same time that PV-INs upregulate Kv3.1, develop a fast-spiking physiological phenotype, and integrate into functional circuits in the third and fourth postnatal weeks. We have updated the discussion (Line 780-782) with this information and more clearly describe our interpretation of these early-life behavioral experiments.

(2) I would like to see a more complete analysis of the Video-EEG data that is included in Figure 8. What was the seizure duration and frequency? Were there spike-wave seizure types observed? Were EEG events that involve thalamocortical circuitry affected such as spindles? Was sleep architecture impaired in the model? Were littermate control animals recorded?

Although classical convulsive seizures represent only part of the overall epilepsy phenotype that this mouse exhibits, we agree that reporting seizure duration and frequency is important. We have now included this in our revised manuscript (line 624-626). We have also now added WT control mice to our dataset, and, as expected, we failed to observe any epileptic features in our WT recordings.

In our EEG experiments, we did not record EMG activity in the mouse to allow for unambiguous determination of sleep vs. quiet wakefulness. For that reason, and because we believe it beyond the scope of this particular study, we did not examine sleep-related EEG phenomena such as spindles or sleep architecture. We have, however, added a line in the discussion (line 771-774) suggesting that future studies focus on a more thorough investigation of the EEG activity in these animals.

(3) The in vivo calcium imaging data shows synchronous bursts in A421V animals which is in agreement with the synchronous bursts observed in the EEG. Overall the analysis of the in vivo calcium imaging data appears to be rudimentary and perhaps this is a missed opportunity. What additional insights were gained from this technically demanding experiment that were not obtained from the EEG recordings?

As noted above, in the revised version of the manuscript, we have conducted additional experiments which allowed us to separately examine PV-IN and non-PV neuron excitability via 2P in vivo calcium imaging. This required an alternative strategy to label individual neuronal somata without contamination by the robust neuropil signal that we observed in the approach undertaken in the original submission. We've described the details of this new approach in methods (Lines 230-271) and results section (lines 630-657).

Our new results (Figure 8 and Supplementary Figure 9) reveal that, during quiet rest, neocortical PV-INs from *Kcnc1-A421V/+* mice exhibit a reduction in calcium transient amplitude during quiet wakefulness and that non-PV cells exhibit altered transient frequency and amplitude. Overall, we believe that these results are consistent with the view that PV-IN-mediated perisomatic inhibition is compromised in *Kcnc1-A421V/+* mice which leads to a downstream hyperexcitability in excitatory neurons within cortical microcircuits.

(4) The increased severity of seizure phenotypes observed in the A421V model relative to knockout mice is interesting but also confusing given what is known about this mutation. As the authors point out, a possible explanation is that the mutation is acting in a dominant negative manner, where mutant Kv3.1 channels compete with other Kvs that would otherwise be able to partially compensate for the loss of Kv function. Alternatively, the A421V mutation might act by affecting the trafficking of heterotetrameric Kv3 channels to the membrane. Can the authors clarify why a trafficking deficit would produce a different effect than a loss of function mutation? Are the authors proposing

that a hypomorphic mutation involving both a partial trafficking deficit and a dominant negative effect of those channels that are properly localized is more severe than a "clean" loss of function? The roughly 50% loss of potassium current absent a change in gating would be expected to behave like a loss-of-function mutation. This might be addressed by comparing the surface expression of the other Kv channels and/or through the use of Kv3.1-selective pharmacology.

These are excellent points raised by the Reviewer. As noted above, we have endeavored to clarify our hypothesis as to the basis of this phenomenon, although the mechanistic basis for the more severe phenotype in the *Kcnc1-A421V/+* mouse relative to the *Kv3.1* knockout is not entirely clear. Our physiology results and the evidence presented supporting a trafficking impairment, are consistent with dominant negative action of the *Kv3.1 A421V* variant at the level of channel gating and/or trafficking. To restate, we think the *Kcnc1-A421V/+* heterozygous variant is more severe than a *Kv3.1* knockout for (at least) three reasons: variant *Kv3.1* is incorporated into *Kv3.1/Kv3.2* heterotetramers to (1) impair trafficking to the membrane as well as (2) alter the electrophysiological function of those channels that do successfully traffic to the membrane (while *Kv3.1* knockout affects *Kv3.1* only), and (3) the heterozygous variant may escape compensatory upregulation of *Kv3.2* and which is known to occur in *Kv3.1* knockout mice.

For example, our data suggests and is consistent with the view that heterotetramers of WT *Kv3.1* and *Kv3.2* potentially come together with the *A421V Kv3.1* subunit in the endoplasmic reticulum and then fail to traffic to the membrane due to the presence of one or more *A421V* subunit(s), as evidenced by increased *Kv3.1* staining in the cytosol in the *Kcnc1-A421V/+* mouse relative to WT. This is in contrast to what would occur in the *Kv3.1* knockout mice as there is no subunit produced from the null allele to impair WT *Kv3.2* subunits from forming fully functional *Kv3.2* homotetramers to then reach the cell surface and function properly. This is one specific possible mechanism for dominant negative activity.

A non-mutually-exclusive mechanism is that inclusion of one or more *Kv3.1 A421V* subunits into *Kv3* heterotetramers impairs gating and prevents potassium flux such that, even if the tetramer does reach the membrane, that entire tetramer fails to contribute to the total potassium current. This is another possible mechanism for dominant negative function of the *A421V* subunit.

Experimental elucidation of the precise mechanism of the dominant negative activity of the *A421V Kcnc1* variant is beyond the scope of this study; yet, our lab is continuing to work on this. It will likely require dose-response experiments in which various ratios of WT and *Kv3.1 A421V* subunits are co-expressed in heterologous cells and then recorded for an overall effect on potassium current similar to (Clatot et al., 2017).

In the revised manuscript, we have updated our discussion of these mechanistic considerations for *KCNC1*-related epilepsy syndromes in lines 868-883 in the Discussion.

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